

From SVMs to Uncertainty-Aware Classification: A Detailed Treatment of Gaussian and Gaussian Mixture Sample Uncertainty

Sushrut Gaikwad

April 20, 2026

Abstract

These notes provide a self-contained, detailed exposition of the Support Vector Machine with Gaussian Sample Uncertainty (SVM-GSU), as proposed by Tzelepis et al. (2017). We begin with the basics of the standard linear SVM and carefully build up to the SVM-GSU formulation, explaining every mathematical step in a way that is accessible to readers with an undergraduate-level background in linear algebra, calculus, and probability. We then develop a novel extension, SVM with Gaussian Mixture Uncertainty (SVM-GMU), which handles non-Gaussian uncertainties by modeling each training example's uncertainty as a mixture of Gaussians. This allows the classifier to handle complex, multimodal, and non-elliptical uncertainty shapes (e.g., banana-shaped contours) while preserving all the desirable properties of the original framework: closed-form loss, closed-form gradients, convexity, and efficient stochastic gradient descent optimization.

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Notation

Throughout this document, we use the following conventions:

- n denotes the number of training examples.
- d denotes the dimensionality of the feature space (i.e., the number of features per example).
- Vectors are **lowercase, bold, non-italic**: $\mathbf{w}$, $\mathbf{x}$, $\mathbf{z}$.
- Greek vectors are **bold upright**: $\boldsymbol{\mu}$, $\boldsymbol{\sigma}$, $\boldsymbol{\theta}$.
- Matrices are **bold upright uppercase**: $\boldsymbol{\Sigma}$, $\boldsymbol{\Lambda}$.
- Scalars are italic: λ , b , d , n .
- In the standard SVM (Part I), $\mathbf{x}_i$ denotes the fixed feature vector of the i -th example.
- In SVM-GSU and SVM-GMU (Parts III and IV), $\boldsymbol{\mu}_i$ denotes the mean of the i -th example's uncertainty distribution, and $\mathbf{x}_i$ denotes a random vector drawn from that distribution: $\mathbf{x}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$.

Part I

Background and the Standard Linear SVM

1 What Is Classification?

Suppose you are a doctor trying to decide whether a tumor is malignant or benign based on some measurements (e.g., size, shape, texture). You have access to historical data, that is, past patients for whom the diagnosis is known. Each patient’s measurements form a *data point*, and the diagnosis (malignant or benign) is the *label*. The goal of *classification* is to learn a rule from this historical data that can predict the label of a new, unseen patient.

More formally, we have a *training set* of n examples:

$$\mathcal{X} = \left\{ (\mathbf{x}_i, y_i) : \mathbf{x}_i \in \mathbb{R}^d, y_i \in \{+1, -1\}, i = 1, \dots, n \right\}, \quad (1)$$

where:

- $\mathbf{x}_i \in \mathbb{R}^d$ is the *feature vector* of the i -th example. It is a d -dimensional column vector. Each entry represents some measured quantity (a “feature”). For example, if $d = 3$, then $\mathbf{x}_i = \begin{bmatrix} x_{i,1} & x_{i,2} & x_{i,3} \end{bmatrix}^\top$ might contain the tumor’s diameter, perimeter, and average intensity.
- $y_i \in \{+1, -1\}$ is the *class label*. We use $+1$ for one class (say, malignant) and -1 for the other (benign).

The classifier’s job is to find a function that, given a new $\mathbf{x}$, predicts whether $y = +1$ or $y = -1$.

2 The Idea of a Linear Classifier

The simplest classifier is a *linear classifier*. In two dimensions, this is just a straight line that separates the two classes. In three dimensions, it is a plane. In d dimensions, it is a *hyperplane*. A hyperplane in $\mathbb{R}^d$ is defined by a weight vector $\mathbf{w} \in \mathbb{R}^d$ and a bias (or intercept) $b \in \mathbb{R}$:

$$H : \quad \mathbf{w}^\top \mathbf{x} + b = 0. \quad (2)$$

Let us unpack this notation:

- $\mathbf{w}^\top \mathbf{x}$ is the *dot product* (or inner product) of $\mathbf{w}$ and $\mathbf{x}$:

$$\mathbf{w}^\top \mathbf{x} = \sum_{j=1}^d w_j x_j = w_1 x_1 + w_2 x_2 + \dots + w_d x_d. \quad (3)$$

This gives a single real number that measures how much $\mathbf{x}$ “aligns” with $\mathbf{w}$.

- b shifts the hyperplane. Without b , the hyperplane would always pass through the origin.

The hyperplane H divides $\mathbb{R}^d$ into two *halfspaces*:

$$\text{Positive side: } \mathbf{w}^\top \mathbf{x} + b > 0, \quad (4)$$

$$\text{Negative side: } \mathbf{w}^\top \mathbf{x} + b < 0. \quad (5)$$

A linear classifier predicts the label of a new point $\mathbf{x}_t$ as:

$$\hat{y}_t = \text{sign}(\mathbf{w}^\top \mathbf{x}_t + b), \quad (6)$$

where “ $\text{sign}(\cdot)$ ” is the sign function: it returns $+1$ if the argument is positive and -1 if negative.

3 The Margin: Why Not Just Any Separating Line?

If the data is *linearly separable* (i.e., a hyperplane exists that perfectly separates the two classes), there are infinitely many such hyperplanes. Which one should we choose? Intuitively, we want the hyperplane that is as far as possible from the nearest data points of both classes. This “buffer zone” is called the *margin*. A larger margin means the classifier is more robust: small perturbations in the data are less likely to cause misclassification.

3.1 Deriving the Distance from a Point to a Hyperplane

Let us derive the distance formula from scratch. Consider a hyperplane $H : \mathbf{w}^\top \mathbf{x} + b = 0$ and a point $\mathbf{x}_0 \in \mathbb{R}^d$. The signed distance from $\mathbf{x}_0$ to H is the length of the shortest vector from $\mathbf{x}_0$ to any point on H .

Let $\mathbf{x}_H$ be the point on H closest to $\mathbf{x}_0$. The vector $\mathbf{x}_0 - \mathbf{x}_H$ must be perpendicular to H . Since $\mathbf{w}$ is the normal vector to H , we have $\mathbf{x}_0 - \mathbf{x}_H = t\mathbf{w}$ for some scalar $t \in \mathbb{R}$. Thus $\mathbf{x}_H = \mathbf{x}_0 - t\mathbf{w}$. Since $\mathbf{x}_H$ lies on H , we have $\mathbf{w}^\top \mathbf{x}_H + b = 0$. Substituting:

$$\mathbf{w}^\top (\mathbf{x}_0 - t\mathbf{w}) + b = 0 \implies \mathbf{w}^\top \mathbf{x}_0 + b - t\|\mathbf{w}\|^2 = 0 \implies t = \frac{\mathbf{w}^\top \mathbf{x}_0 + b}{\|\mathbf{w}\|^2}. \quad (7)$$

The distance is $\|\mathbf{x}_0 - \mathbf{x}_H\| = |t| \cdot \|\mathbf{w}\|$:

$$\text{distance}(\mathbf{x}_0, H) = \frac{|\mathbf{w}^\top \mathbf{x}_0 + b|}{\|\mathbf{w}\|}, \quad (8)$$

where $\|\mathbf{w}\| = \sqrt{\mathbf{w}^\top \mathbf{w}} = \sqrt{\sum_j w_j^2}$ is the Euclidean norm (length) of $\mathbf{w}$. The Support Vector Machine (SVM) is the classifier that *maximizes* this margin.

4 The SVM Optimization Problem

In practice, data may not be perfectly separable. The standard *soft-margin SVM* allows some misclassifications but penalizes them. The optimization problem is:

$$\min_{\mathbf{w}, b} \mathcal{J}_{\text{SVM}}(\mathbf{w}, b) = \underbrace{\frac{\lambda}{2} \|\mathbf{w}\|^2}_{\text{regularization}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))}_{\text{average hinge loss over training set}}. \quad (9)$$

Let us understand each term.

4.1 The Regularization Term: $\frac{\lambda}{2} \|\mathbf{w}\|^2$

This is the *regularization* term. Here $\lambda > 0$ is a user-chosen parameter that controls how much we care about keeping $\|\mathbf{w}\|$ small (equivalently, keeping the margin large, since the geometric margin equals $1/\|\mathbf{w}\|$). A larger λ means we prioritize a wider margin even if it means more misclassifications.

4.2 The Hinge Loss: $\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$

This is the *hinge loss*, often written as $h(t) = \max(0, 1 - t)$ where $t = y_i(\mathbf{w}^\top \mathbf{x}_i + b)$. The quantity $t_i = y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ is called the *functional margin* of point i . Let us see what happens:

- **Correctly classified and far from boundary** ($t_i \geq 1$): The hinge loss is zero. The point contributes no penalty. These are the “easy” points.
- **Correctly classified but close to boundary** ($0 < t_i < 1$): The hinge loss is $1 - t_i > 0$. The point is correctly classified but falls within the margin. It is penalized mildly.
- **Misclassified** ($t_i < 0$): The hinge loss is $1 - t_i > 1$. The point is on the wrong side of the boundary and is penalized heavily. The further away it is on the wrong side, the larger the penalty.

The SVM objective eq. (9) balances two competing goals:

1. **Wide margin** (small $\|\mathbf{w}\|$): Controlled by the regularization term.
2. **Few misclassifications** (small hinge loss): Controlled by the data-fitting term.

The parameter λ trades off between these two goals.

4.3 A Note on Convexity

The SVM objective eq. (9) is a *convex function*. This is a crucial property. A function is convex if, roughly speaking, it is “bowl-shaped”, that is, it has no local minima other than the global minimum. This means:

- Any local minimum is also the global minimum.
- We can find the optimal $\mathbf{w}$ and b efficiently using gradient-based optimization.

Convexity follows because $\|\mathbf{w}\|^2$ is convex, $\max(0, 1 - t)$ is convex in t (it is piecewise linear), and $t = y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ is linear in $(\mathbf{w}, b)$. A sum of convex functions is convex.

Part II

The Problem: What If Our Data Is Uncertain?

5 Why Data Uncertainty Matters

In the standard SVM formulation above, each training example $\mathbf{x}_i$ is treated as a *fixed, known point* in $\mathbb{R}^d$. But in practice, this is often not realistic:

1. **Measurement noise:** Every measurement instrument has some imprecision. The true value is not exactly $\mathbf{x}_i$ but something “close to” $\mathbf{x}_i$, with some uncertainty.
2. **Feature extraction artifacts:** In computer vision, natural language processing, or audio processing, the features are computed via complex pipelines. The extracted features are noisy approximations of the underlying signal.

3. **Aggregated data:** Sometimes $\mathbf{x}_i$ is the *average* of many measurements. For example, in a medical dataset, the reported value might be the average of multiple cell measurements, and the standard deviation is also available.
4. **Data augmentation modeling:** An image can be shifted, rotated, or scaled by small amounts. The set of images produced by these transforms can be modeled as a distribution centered at the original image.

In all these cases, we know (or can estimate) not just a single point $\mathbf{x}_i$, but a *probability distribution* around $\mathbf{x}_i$ that describes where the “true” data point might lie.

6 Why the Standard SVM Fails to Use This Information

The standard SVM takes only the point $\mathbf{x}_i$ (typically the mean) and ignores any associated uncertainty. Consider two scenarios:

- **Scenario A:** A data point $\mathbf{x}_i$ lies very close to the decision boundary, but we are very certain about its position (tiny uncertainty).
- **Scenario B:** The same point $\mathbf{x}_i$ lies close to the boundary, but we are very *uncertain* about its position (large uncertainty that extends across the boundary).

The standard SVM treats these two scenarios identically. But clearly, in Scenario B, there is a significant probability that the true data point actually lies on the wrong side of the boundary. A good classifier should account for this.

Part III

SVM with Gaussian Sample Uncertainty (SVM-GSU)

This part presents the method proposed by Tzelepis et al. (2017). Each training example is modeled not as a fixed point, but as a Gaussian distribution.

Roadmap for Part III

We will proceed in five steps:

1. **Define** the new training set, replacing fixed points with Gaussians (section 7).
2. **Write down** the objective function containing an integral we cannot yet evaluate (section 8).
3. **Evaluate** that integral in closed form using a key theorem (section 9).
4. **Substitute** the result back to obtain the full, explicit objective function.
5. **Differentiate** the objective to get closed-form gradients for optimization (section 10).

At each stage, we will display the objective function so that you always see the bigger picture.

7 The New Training Set

In the standard SVM, the i -th training example was a fixed vector $\mathbf{x}_i$. Now, we make a conceptual shift: what was previously a fixed point is now replaced by a *probability distribution* describing where the true data point might lie. Specifically, we model each training example as a Gaussian distribution with a known mean and covariance.

To make this shift clear in our notation, we use $\boldsymbol{\mu}_i$ to denote the *mean* (center) of the i -th example's distribution, and $\mathbf{x}_i$ to denote a *random vector* drawn from that distribution. The training set becomes:

$$\mathcal{X}' = \left\{ (\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, y_i) : \boldsymbol{\mu}_i \in \mathbb{R}^d, \boldsymbol{\Sigma}_i \in \mathbb{S}_{++}^d, y_i \in \{+1, -1\}, i = 1, \dots, n \right\}, \quad (10)$$

where:

- $\boldsymbol{\mu}_i \in \mathbb{R}^d$ is the *mean vector* of the i -th example. This is the “center” of the uncertainty distribution, that is, the best estimate of where the data point lies.
- $\boldsymbol{\Sigma}_i \in \mathbb{S}_{++}^d$ is the *covariance matrix* of the i -th example. It is a $d \times d$ symmetric positive definite matrix that describes the shape and extent of the uncertainty. Here, $\mathbb{S}_{++}^d$ denotes the set of all $d \times d$ symmetric positive definite matrices.
- $y_i \in \{+1, -1\}$ is the class label, as before.

We then define a random vector $\mathbf{x}_i$ for each training example, and assume it follows a multivariate Gaussian (normal) distribution:

$$\mathbf{x}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i). \quad (11)$$

This notation reads naturally: “the i -th data point $\mathbf{x}_i$ is drawn from a Gaussian with mean $\boldsymbol{\mu}_i$ and covariance $\boldsymbol{\Sigma}_i$.” The mean $\boldsymbol{\mu}_i$ plays the role that the fixed point $\mathbf{x}_i$ played in the standard SVM, but now we also have the covariance $\boldsymbol{\Sigma}_i$ encoding the uncertainty around it.

7.1 What Is a Multivariate Gaussian Distribution?

For readers who may be less familiar, let us briefly review this concept. A random vector $\mathbf{x} \in \mathbb{R}^d$ follows a multivariate Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ if its probability density function (PDF) is:

$$f_{\mathbf{x}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad (12)$$

where:

- $|\boldsymbol{\Sigma}|$ is the determinant of $\boldsymbol{\Sigma}$.
- $\boldsymbol{\Sigma}^{-1}$ is the inverse of $\boldsymbol{\Sigma}$.
- The quantity $(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$ is called the *Mahalanobis distance* squared. It measures how far $\mathbf{x}$ is from the mean $\boldsymbol{\mu}$, taking into account the shape of the distribution.

The *iso-density contours* (curves where $f_{\mathbf{x}}$ is constant) are *ellipsoids* centered at $\boldsymbol{\mu}$. The shape and orientation of these ellipsoids are determined by $\boldsymbol{\Sigma}$:

- If $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ (a scalar times the identity matrix), the contours are *spheres*. The uncertainty is the same in every direction. This is called *isotropic* uncertainty.

- If Σ is diagonal (but with different diagonal entries), the contours are axis-aligned ellipsoids. The uncertainty differs along the coordinate axes.
- If Σ is a general positive definite matrix, the contours are arbitrarily oriented ellipsoids. This is *anisotropic* uncertainty.

7.2 A Useful Property: Linear Transformations of Gaussians

We will need the following standard fact. If $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ and $\mathbf{a} \in \mathbb{R}^d$ is a fixed vector, then the scalar $\mathbf{a}^\top \mathbf{x}$ follows a univariate Gaussian:

$$\mathbf{a}^\top \mathbf{x} \sim \mathcal{N}(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \Sigma \mathbf{a}). \quad (13)$$

Proof. The mean is $\mathbb{E}[\mathbf{a}^\top \mathbf{x}] = \mathbf{a}^\top \mathbb{E}[\mathbf{x}] = \mathbf{a}^\top \boldsymbol{\mu}$. The variance is

$$\text{Var}(\mathbf{a}^\top \mathbf{x}) = \mathbb{E}[(\mathbf{a}^\top \mathbf{x} - \mathbf{a}^\top \boldsymbol{\mu})^2] = \mathbb{E}[\mathbf{a}^\top (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{a}] = \mathbf{a}^\top \Sigma \mathbf{a}. \quad (14)$$

Gaussianity is preserved under linear transformations (a well-known property of the Gaussian family).

8 The Key Idea: Expected Hinge Loss

Now comes the central idea of SVM-GSU. Instead of computing the hinge loss at the fixed point $\boldsymbol{\mu}_i$ (as the standard SVM would), we compute the *expected value* (average) of the hinge loss over all possible positions of the i -th training example, weighted by the probability of each position under the Gaussian $\mathcal{N}(\boldsymbol{\mu}_i, \Sigma_i)$.

8.1 Why Take the Expected Value?

Think of it this way: the i -th training example is not a single point but a “cloud” of possible points, described by the Gaussian $\mathcal{N}(\boldsymbol{\mu}_i, \Sigma_i)$. Each possible point $\mathbf{x}$ in this cloud would incur a hinge loss of $\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b))$. The expected hinge loss averages this over the entire cloud, weighting each point by how likely it is.

Points that are more probable (near the mean $\boldsymbol{\mu}_i$) contribute more to the expected loss. Points that are less probable (far from $\boldsymbol{\mu}_i$) contribute less. This is exactly the right thing to do if we believe the data comes from this distribution.

8.2 The SVM-GSU Objective Function (Integral Form)

Recall that the standard SVM objective (eq. (9)) was:

$$\mathcal{J}_{\text{SVM}}(\mathbf{w}, b) = \underbrace{\frac{\lambda}{2} \|\mathbf{w}\|^2}_{\text{regularization}} + \frac{1}{n} \sum_{i=1}^n \underbrace{\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))}_{\text{hinge loss at } \mathbf{x}_i}.$$

The SVM-GSU objective replaces the hinge loss at the fixed point $\mathbf{x}_i$ with the *expected* hinge loss over the Gaussian $\mathcal{N}(\boldsymbol{\mu}_i, \Sigma_i)$. The regularization term stays the same:

$$\mathcal{J}_{\text{GSU}}(\mathbf{w}, b) = \underbrace{\frac{\lambda}{2} \|\mathbf{w}\|^2}_{\text{regularization}} + \frac{1}{n} \sum_{i=1}^n \underbrace{\int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}}_{\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b) = \text{expected hinge loss for example } i}, \quad (15)$$

where $f_{\mathbf{x}_i}(\mathbf{x})$ is the Gaussian PDF of the i -th example (eq. (12) with $\boldsymbol{\mu} = \boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_i$). Note that $\mathbf{x}$ in the integral is a *dummy variable of integration*. It ranges over all possible positions in $\mathbb{R}^d$.

We can write the loss for the i -th example more compactly using expectation notation:

$$\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b) = \mathbb{E}_{\mathbf{x}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}[\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))]. \quad (16)$$

SVM-GSU Objective - Integral Form (eq. (15))

$$\mathcal{J}_{\text{GSU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_{i=1}^n \underbrace{\int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}}_{\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b) = \text{expected hinge loss for example } i}.$$

The unknowns are $\mathbf{w}$ and b . Everything else ($\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, y_i$) is given data. The integral $\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b)$ is not yet in a usable form. We evaluate it next.

9 Evaluating the Loss in Closed Form

The integral $\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b)$ in eq. (15) looks daunting. Remarkably, it can be evaluated in *closed form*, meaning we can write down an explicit formula without needing to compute the integral numerically.

9.1 Splitting the Integral Over a Halfspace

The hinge loss $\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b))$ is zero when $y_i(\mathbf{w}^\top \mathbf{x} + b) \geq 1$, and equals $1 - y_i(\mathbf{w}^\top \mathbf{x} + b)$ when $y_i(\mathbf{w}^\top \mathbf{x} + b) < 1$. Therefore, the integral is only nonzero over the *halfspace*:

$$\Omega_i = \{\mathbf{x} \in \mathbb{R}^d : y_i(\mathbf{w}^\top \mathbf{x} + b) \leq 1\}. \quad (17)$$

This is the set of points that are either misclassified or lie within the margin. So the loss becomes:

$$\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b) = \int_{\Omega_i} [1 - y_i(\mathbf{w}^\top \mathbf{x} + b)] f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}. \quad (18)$$

This integral involves integrating a *linear function* times a *Gaussian PDF* over a *halfspace*. This is a well-known type of integral that can be evaluated analytically.

9.2 The Closed-Form Result (Theorem 1 from the Paper)

The paper proves (in Appendix A) the following key result. We state it here, and provide a full proof in Appendix A.

Theorem 9.1 (Gaussian-Weighted Linear Integral Over a Halfspace). *Let $\mathbf{x} \in \mathbb{R}^d$ be a random vector with $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Let H be the hyperplane $\mathbf{a}^\top \mathbf{x} + b' = 0$, which divides $\mathbb{R}^d$ into two halfspaces,*

$$\Omega^\pm = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}^\top \mathbf{x} + b' \gtrless 0\}. \quad (19)$$

Then the integrals

$$I^\pm(\mathbf{a}, b') \equiv \int_{\Omega^\pm} (\mathbf{a}^\top \mathbf{x} + b') f_{\mathbf{x}}(\mathbf{x}) \, d\mathbf{x} \quad (20)$$

evaluate to

$$I^\pm(\mathbf{a}, b') = \frac{d_{\boldsymbol{\mu}}}{2} \left(1 \pm \operatorname{erf}\left(\frac{d_{\boldsymbol{\mu}}}{d_{\boldsymbol{\Sigma}}}\right) \right) \pm \frac{d_{\boldsymbol{\Sigma}}}{2\sqrt{\pi}} \exp\left(-\frac{d_{\boldsymbol{\mu}}^2}{d_{\boldsymbol{\Sigma}}^2}\right), \quad (21)$$

where

$$d_{\boldsymbol{\mu}} = \mathbf{a}^\top \boldsymbol{\mu} + b', \quad (22)$$

$$d_{\boldsymbol{\Sigma}} = \sqrt{2 \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}}, \quad (23)$$

and $\text{erf}(\cdot)$ is the error function, defined as:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt. \quad (24)$$

9.3 What Is the Error Function?

The error function $\text{erf}(x)$ is closely related to the cumulative distribution function (CDF) of the standard normal distribution in the following way:

$$\Phi(x) = \frac{1}{2} \left(1 + \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right).$$

This error function maps $\mathbb{R}$ to $(-1, 1)$, with the following properties:

- $\text{erf}(0) = 0$.
- $\lim_{x \rightarrow \infty} \text{erf}(x) = 1$.
- $\lim_{x \rightarrow -\infty} \text{erf}(x) = -1$.
- It is an odd function: $\text{erf}(-x) = -\text{erf}(x)$.

It appears naturally whenever we integrate Gaussian functions over half-lines or halfspaces. It is available as a standard function in essentially all numerical computing libraries.

An important identity we will use repeatedly is the derivative of the error function:

$$\frac{d}{dx} \text{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}. \quad (25)$$

This follows directly from eq. (24) by the fundamental theorem of calculus.

9.4 Applying the Theorem to Obtain the SVM-GSU Loss

To apply theorem 9.1 to our loss function eq. (18), we need to match the notation. The halfspace Ω_i is defined by $y_i(\mathbf{w}^\top \mathbf{x} + b) \leq 1$. Rearranging:

$$y_i(\mathbf{w}^\top \mathbf{x} + b) \leq 1 \iff y_i \mathbf{w}^\top \mathbf{x} + y_i b - 1 \leq 0 \iff -y_i \mathbf{w}^\top \mathbf{x} + (1 - y_i b) \geq 0. \quad (26)$$

The integrand is $[1 - y_i(\mathbf{w}^\top \mathbf{x} + b)]$, which equals $(-y_i \mathbf{w})^\top \mathbf{x} + (1 - y_i b)$. Comparing with the theorem's integrand $\mathbf{a}^\top \mathbf{x} + b'$ over the halfspace $\Omega^+ = \{\mathbf{x} : \mathbf{a}^\top \mathbf{x} + b' \geq 0\}$, we identify:

$$\mathbf{a} = -y_i \mathbf{w}, \quad b' = 1 - y_i b. \quad (27)$$

Substituting into eqs. (22) and (23):

$$\begin{aligned} d_{\boldsymbol{\mu}_i} &\equiv d_{\boldsymbol{\mu}} = (-y_i \mathbf{w})^\top \boldsymbol{\mu}_i + (1 - y_i b) = -y_i \mathbf{w}^\top \boldsymbol{\mu}_i + 1 - y_i b \\ &= 1 - y_i(\mathbf{w}^\top \boldsymbol{\mu}_i + b), \end{aligned} \quad (28)$$

$$\begin{aligned} d_{\boldsymbol{\Sigma}_i} &\equiv d_{\boldsymbol{\Sigma}} = \sqrt{2 (-y_i \mathbf{w})^\top \boldsymbol{\Sigma}_i (-y_i \mathbf{w})} = \sqrt{2 y_i^2 \mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}} \\ &= \sqrt{2 \mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}}, \end{aligned} \quad (29)$$

where in the last step we used $y_i^2 = 1$ (since $y_i \in \{+1, -1\}$). Notice that d_{Σ_i} does not depend on y_i or on b . Applying the “+” branch of eq. (21), the closed-form loss for the i -th training example is:

$$\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b) = \frac{d_{\mu_i}}{2} \left[\text{erf}\left(\frac{d_{\mu_i}}{d_{\Sigma_i}}\right) + 1 \right] + \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \exp\left(-\frac{d_{\mu_i}^2}{d_{\Sigma_i}^2}\right). \quad (30)$$

9.5 The Full Objective with the Closed-Form Loss

Now we substitute eq. (30) back into the objective eq. (15). The integral has been eliminated and the objective is now a fully explicit function of $\mathbf{w}$ and b :

$$\mathcal{J}_{\text{GSU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_{i=1}^n \underbrace{\left[\frac{d_{\mu_i}}{2} \left(\text{erf}\left(\frac{d_{\mu_i}}{d_{\Sigma_i}}\right) + 1 \right) + \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \exp\left(-\frac{d_{\mu_i}^2}{d_{\Sigma_i}^2}\right) \right]}_{\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b)}, \quad (31)$$

where $d_{\mu_i} = 1 - y_i(\mathbf{w}^\top \mu_i + b)$ and $d_{\Sigma_i} = \sqrt{2 \mathbf{w}^\top \Sigma_i \mathbf{w}}$.

SVM-GSU Objective - Closed Form (eq. (31))

$$\mathcal{J}_{\text{GSU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_{i=1}^n \underbrace{\left[\frac{d_{\mu_i}}{2} \left(\text{erf}\left(\frac{d_{\mu_i}}{d_{\Sigma_i}}\right) + 1 \right) + \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \exp\left(-\frac{d_{\mu_i}^2}{d_{\Sigma_i}^2}\right) \right]}_{\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b)}$$

The only unknowns are $\mathbf{w}$ and b . The quantities d_{μ_i} and d_{Σ_i} are shorthand that depend on $\mathbf{w}$, b , and the given data (μ_i, Σ_i, y_i) . No integrals remain. This is what we minimize.

9.6 Understanding the Two Quantities

Let us build intuition for the two key quantities d_{μ_i} and d_{Σ_i} appearing in the objective:

- $d_{\mu_i} = 1 - y_i(\mathbf{w}^\top \mu_i + b)$:
 - This is exactly the hinge loss argument of the *mean* μ_i in the standard SVM. If the mean is correctly classified and lies outside the margin, $d_{\mu_i} < 0$. If the mean lies inside the margin or is misclassified, $d_{\mu_i} > 0$.
- $d_{\Sigma_i} = \sqrt{2 \mathbf{w}^\top \Sigma_i \mathbf{w}}$:
 - This measures the *spread of the uncertainty in the direction of $\mathbf{w}$* . Using the linear-transformation property discussed in section 7.2, if $\mathbf{x}_i \sim \mathcal{N}(\mu_i, \Sigma_i)$, then $\mathbf{w}^\top \mathbf{x}_i \sim \mathcal{N}(\mathbf{w}^\top \mu_i, \mathbf{w}^\top \Sigma_i \mathbf{w})$. So $\mathbf{w}^\top \Sigma_i \mathbf{w}$ is exactly the variance of the projection of $\mathbf{x}_i$ onto the direction $\mathbf{w}$, and d_{Σ_i} is $\sqrt{2}$ times the standard deviation. This captures how much the uncertainty “matters” for classification, specifically, how much the Gaussian is spread out in the direction perpendicular to the decision boundary (since $\mathbf{w}$ is the normal vector to the hyperplane).

Remark 9.1 (Sanity check). Look at eq. (31). The first term is the regularizer, identical to standard SVM. The second term is a sum over training examples, where each contribution depends on two things: how far the mean μ_i is from the decision boundary (via d_{μ_i}) and how spread out the uncertainty is in the classification-relevant direction (via d_{Σ_i}). If a training example has large uncertainty that straddles the boundary, its loss contribution will be large, even if its mean is on the correct side.

9.7 Limiting Behavior: Recovering Standard SVM

What happens in eq. (31) when the uncertainty vanishes, i.e., $\Sigma_i \rightarrow \mathbf{0}$? Let us verify that SVM-GSU recovers the standard SVM hinge loss in this limit. As $\Sigma_i \rightarrow \mathbf{0}$, we have $d_{\Sigma_i} = \sqrt{2\mathbf{w}^\top \Sigma_i \mathbf{w}} \rightarrow 0^+$.

Case 1: $d_{\mu_i} > 0$ (mean is inside margin or misclassified). Then $\frac{d_{\mu_i}}{d_{\Sigma_i}} \rightarrow +\infty$, so we get $\text{erf}\left(\frac{d_{\mu_i}}{d_{\Sigma_i}}\right) \rightarrow 1$. The first term becomes

$$\frac{d_{\mu_i}}{2}(1+1) = d_{\mu_i}. \quad (32)$$

For the second term, $\exp\left(-\frac{d_{\mu_i}^2}{d_{\Sigma_i}^2}\right) \rightarrow 0$ faster than any polynomial, and $d_{\Sigma_i} \rightarrow 0$, so the product $\frac{d_{\Sigma_i}}{2\sqrt{\pi}} \exp\left(-\frac{d_{\mu_i}^2}{d_{\Sigma_i}^2}\right) \rightarrow 0$. Thus $\mathcal{L}_{\text{GSU}i} \rightarrow d_{\mu_i} = 1 - y_i(\mathbf{w}^\top \mu_i + b) > 0$.

Case 2: $d_{\mu_i} < 0$ (mean is correctly classified, outside margin). Then $\frac{d_{\mu_i}}{d_{\Sigma_i}} \rightarrow -\infty$, so $\text{erf}\left(\frac{d_{\mu_i}}{d_{\Sigma_i}}\right) \rightarrow -1$. The first term becomes

$$\frac{d_{\mu_i}}{2}(-1+1) = 0. \quad (33)$$

The second term again vanishes. Thus $\mathcal{L}_{\text{GSU}i} \rightarrow 0$.

Conclusion. In both cases, $\mathcal{L}_{\text{GSU}i} \rightarrow \max(0, d_{\mu_i}) = \max(0, 1 - y_i(\mathbf{w}^\top \mu_i + b))$, which is exactly the standard SVM hinge loss evaluated at the mean. This confirms that **SVM-GSU is a strict generalization of the standard SVM**. The standard SVM is recovered as a special case when all covariance matrices tend to zero.

10 Gradients in Closed Form

To minimize the objective $\mathcal{J}_{\text{GSU}}(\mathbf{w}, b)$ (i.e., eq. (31)) using gradient descent, we need its partial derivatives with respect to $\mathbf{w}$ and b . We derive them below in full detail.

10.1 Setup

We restate the objective:

$$\mathcal{J}_{\text{GSU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_{i=1}^n \underbrace{\left[\frac{d_{\mu_i}}{2} \left(\text{erf}\left(\frac{d_{\mu_i}}{d_{\Sigma_i}}\right) + 1 \right) + \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \exp\left(-\frac{d_{\mu_i}^2}{d_{\Sigma_i}^2}\right) \right]}_{\mathcal{L}_{\text{GSU}i}(\mathbf{w}, b)}, \quad (34)$$

where

$$\begin{aligned} d_{\mu_i} &= 1 - y_i(\mathbf{w}^\top \mu_i + b), \\ d_{\Sigma_i} &= \sqrt{2\mathbf{w}^\top \Sigma_i \mathbf{w}}. \end{aligned}$$

We observe:

- $\mathcal{L}_{\text{GSU}_i}$ depends on $\mathbf{w}$ through *both* d_{μ_i} (linearly, via $\mathbf{w}^\top \mu_i$) and d_{Σ_i} (via the quadratic form $\mathbf{w}^\top \Sigma_i \mathbf{w}$).
- $\mathcal{L}_{\text{GSU}_i}$ depends on b only through d_{μ_i} .

We will therefore use the chain rule. Let $r_i \equiv \frac{d_{\mu_i}}{d_{\Sigma_i}}$ for brevity.

10.2 Step 1: Compute $\frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\mu_i}}$

We differentiate eq. (30) with respect to d_{μ_i} , treating d_{Σ_i} as a constant (the dependence of d_{Σ_i} on $\mathbf{w}$ will be handled separately). The loss is

$$\mathcal{L}_{\text{GSU}_i} = \underbrace{\frac{d_{\mu_i}}{2} (\text{erf}(r_i) + 1)}_{\text{term A}} + \underbrace{\frac{d_{\Sigma_i}}{2\sqrt{\pi}} \exp(-r_i^2)}_{\text{term B}}. \quad (35)$$

Derivative of term A. By the product rule, treating $r_i = \frac{d_{\mu_i}}{d_{\Sigma_i}}$:

$$\begin{aligned} \frac{\partial A}{\partial d_{\mu_i}} &= \frac{1}{2}(\text{erf}(r_i) + 1) + \frac{d_{\mu_i}}{2} \cdot \frac{\partial}{\partial d_{\mu_i}} \text{erf}(r_i) \\ &= \frac{1}{2}(\text{erf}(r_i) + 1) + \frac{d_{\mu_i}}{2} \cdot \text{erf}'(r_i) \cdot \frac{\partial r_i}{\partial d_{\mu_i}}. \end{aligned} \quad (36)$$

Using eq. (25), $\text{erf}'(r_i) = \frac{2}{\sqrt{\pi}} e^{-r_i^2}$, and $\frac{\partial r_i}{\partial d_{\mu_i}} = \frac{1}{d_{\Sigma_i}}$:

$$\frac{\partial A}{\partial d_{\mu_i}} = \frac{1}{2}(\text{erf}(r_i) + 1) + \frac{d_{\mu_i}}{2} \cdot \frac{2}{\sqrt{\pi}} e^{-r_i^2} \cdot \frac{1}{d_{\Sigma_i}} = \frac{1}{2}(\text{erf}(r_i) + 1) + \frac{d_{\mu_i} e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}}. \quad (37)$$

Derivative of term B. Here d_{Σ_i} is treated as constant and only r_i depends on d_{μ_i} :

$$\frac{\partial B}{\partial d_{\mu_i}} = \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \cdot e^{-r_i^2} \cdot (-2r_i) \cdot \frac{\partial r_i}{\partial d_{\mu_i}} = \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \cdot e^{-r_i^2} \cdot (-2r_i) \cdot \frac{1}{d_{\Sigma_i}} = -\frac{r_i e^{-r_i^2}}{\sqrt{\pi}}. \quad (38)$$

Substituting $r_i = \frac{d_{\mu_i}}{d_{\Sigma_i}}$:

$$\frac{\partial B}{\partial d_{\mu_i}} = -\frac{d_{\mu_i} e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}}. \quad (39)$$

Sum. Adding the two derivatives, the terms $\frac{d_{\mu_i} e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}}$ and $-\frac{d_{\mu_i} e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}}$ cancel:

$$\boxed{\frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\mu_i}} = \frac{1}{2}(\text{erf}(r_i) + 1).} \quad (40)$$

This is a remarkably clean result.

10.3 Step 2: Compute $\frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\Sigma_i}}$

Now we differentiate with respect to d_{Σ_i} , treating d_{μ_i} as constant.

Derivative of term A. Only r_i depends on d_{Σ_i} here, since the coefficient $\frac{d_{\mu_i}}{2}$ is constant. Using $\frac{\partial r_i}{\partial d_{\Sigma_i}} = -\frac{d_{\mu_i}}{d_{\Sigma_i}^2}$:

$$\frac{\partial A}{\partial d_{\Sigma_i}} = \frac{d_{\mu_i}}{2} \cdot \text{erf}'(r_i) \cdot \frac{\partial r_i}{\partial d_{\Sigma_i}} = \frac{d_{\mu_i}}{2} \cdot \frac{2}{\sqrt{\pi}} e^{-r_i^2} \cdot \left(-\frac{d_{\mu_i}}{d_{\Sigma_i}^2} \right) = -\frac{d_{\mu_i}^2 e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}^2}. \quad (41)$$

Derivative of term B. By the product rule:

$$\begin{aligned} \frac{\partial B}{\partial d_{\Sigma_i}} &= \frac{1}{2\sqrt{\pi}} e^{-r_i^2} + \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \cdot e^{-r_i^2} \cdot (-2r_i) \cdot \frac{\partial r_i}{\partial d_{\Sigma_i}} \\ &= \frac{e^{-r_i^2}}{2\sqrt{\pi}} + \frac{d_{\Sigma_i}}{2\sqrt{\pi}} \cdot e^{-r_i^2} \cdot (-2r_i) \cdot \left(-\frac{d_{\mu_i}}{d_{\Sigma_i}^2} \right) \\ &= \frac{e^{-r_i^2}}{2\sqrt{\pi}} + \frac{r_i d_{\mu_i} e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}} \\ &= \frac{e^{-r_i^2}}{2\sqrt{\pi}} + \frac{d_{\mu_i}^2 e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}^2}, \end{aligned} \quad (42)$$

where in the last step we used $r_i = \frac{d_{\mu_i}}{d_{\Sigma_i}}$.

Sum. Adding, the terms $-\frac{d_{\mu_i}^2 e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}^2}$ and $+\frac{d_{\mu_i}^2 e^{-r_i^2}}{\sqrt{\pi} d_{\Sigma_i}^2}$ cancel:

$$\boxed{\frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\Sigma_i}} = \frac{e^{-r_i^2}}{2\sqrt{\pi}}.} \quad (43)$$

10.4 Step 3: Compute $\frac{\partial d_{\mu_i}}{\partial \mathbf{w}}$ and $\frac{\partial d_{\mu_i}}{\partial b}$

Since $d_{\mu_i} = 1 - y_i(\mathbf{w}^\top \boldsymbol{\mu}_i + b) = 1 - y_i \mathbf{w}^\top \boldsymbol{\mu}_i - y_i b$:

$$\frac{\partial d_{\mu_i}}{\partial \mathbf{w}} = -y_i \boldsymbol{\mu}_i, \quad (44)$$

$$\frac{\partial d_{\mu_i}}{\partial b} = -y_i. \quad (45)$$

10.5 Step 4: Compute $\frac{\partial d_{\Sigma_i}}{\partial \mathbf{w}}$

We have $d_{\Sigma_i} = \sqrt{2\mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}}$. First, note that for a symmetric matrix $\boldsymbol{\Sigma}_i$,

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}) = (\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_i^\top) \mathbf{w} = 2\boldsymbol{\Sigma}_i \mathbf{w}. \quad (46)$$

Therefore, by the chain rule:

$$\frac{\partial d_{\Sigma_i}}{\partial \mathbf{w}} = \frac{1}{2} (2\mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w})^{-1/2} \cdot 2 \cdot 2\boldsymbol{\Sigma}_i \mathbf{w} = \frac{2\boldsymbol{\Sigma}_i \mathbf{w}}{\sqrt{2\mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}}} = \frac{2\boldsymbol{\Sigma}_i \mathbf{w}}{d_{\Sigma_i}}. \quad (47)$$

Also, d_{Σ_i} does not depend on b :

$$\frac{\partial d_{\Sigma_i}}{\partial b} = \mathbf{0}. \quad (48)$$

10.6 Step 5: Assemble the Gradient with Respect to $\mathbf{w}$

By the chain rule:

$$\frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial \mathbf{w}} = \frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\boldsymbol{\mu}_i}} \cdot \frac{\partial d_{\boldsymbol{\mu}_i}}{\partial \mathbf{w}} + \frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\boldsymbol{\Sigma}_i}} \cdot \frac{\partial d_{\boldsymbol{\Sigma}_i}}{\partial \mathbf{w}}. \quad (49)$$

Substituting eqs. (40), (43), (44) and (47):

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial \mathbf{w}} &= \frac{1}{2}(\text{erf}(r_i) + 1) \cdot (-y_i \boldsymbol{\mu}_i) + \frac{e^{-r_i^2}}{2\sqrt{\pi}} \cdot \frac{2\boldsymbol{\Sigma}_i \mathbf{w}}{d_{\boldsymbol{\Sigma}_i}} \\ &= \frac{e^{-r_i^2}}{\sqrt{\pi} d_{\boldsymbol{\Sigma}_i}} \boldsymbol{\Sigma}_i \mathbf{w} - \frac{y_i}{2} [\text{erf}(r_i) + 1] \boldsymbol{\mu}_i. \end{aligned} \quad (50)$$

Substituting back $r_i = \frac{d_{\boldsymbol{\mu}_i}}{d_{\boldsymbol{\Sigma}_i}}$ and adding the regularizer's gradient $\lambda \mathbf{w}$:

$$\boxed{\frac{\partial \mathcal{J}_{\text{GSU}}}{\partial \mathbf{w}} = \lambda \mathbf{w} + \frac{1}{n} \sum_{i=1}^n \left[\frac{\exp\left(-\frac{d_{\boldsymbol{\mu}_i}^2}{d_{\boldsymbol{\Sigma}_i}^2}\right)}{\sqrt{\pi} d_{\boldsymbol{\Sigma}_i}} \boldsymbol{\Sigma}_i \mathbf{w} - \frac{y_i}{2} \left(\text{erf}\left(\frac{d_{\boldsymbol{\mu}_i}}{d_{\boldsymbol{\Sigma}_i}}\right) + 1 \right) \boldsymbol{\mu}_i \right]}. \quad (51)$$

The two terms inside the sum have clear interpretations:

1. The first term, involving $\boldsymbol{\Sigma}_i \mathbf{w}$, arises from the dependence of $d_{\boldsymbol{\Sigma}_i}$ on $\mathbf{w}$. It has no analog in the standard SVM and is the new contribution due to uncertainty.
2. The second term, involving $\boldsymbol{\mu}_i$, is analogous to the gradient of the hinge loss in standard SVM, but modulated by a “soft” factor $\frac{1}{2}(\text{erf}(\cdot) + 1)$ that smoothly transitions between 0 and 1 instead of the hard step function in standard SVM.

10.7 Step 6: Assemble the Gradient with Respect to b

Since $d_{\boldsymbol{\Sigma}_i}$ does not depend on b (eq. (48)), only the $d_{\boldsymbol{\mu}_i}$ pathway contributes:

$$\frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial b} = \frac{\partial \mathcal{L}_{\text{GSU}_i}}{\partial d_{\boldsymbol{\mu}_i}} \cdot \frac{\partial d_{\boldsymbol{\mu}_i}}{\partial b} = \frac{1}{2}(\text{erf}(r_i) + 1) \cdot (-y_i) = -\frac{y_i}{2}(\text{erf}(r_i) + 1). \quad (52)$$

Summing over all examples (the regularizer does not depend on b):

$$\boxed{\frac{\partial \mathcal{J}_{\text{GSU}}}{\partial b} = -\frac{1}{n} \sum_{i=1}^n \frac{y_i}{2} \left[\text{erf}\left(\frac{d_{\boldsymbol{\mu}_i}}{d_{\boldsymbol{\Sigma}_i}}\right) + 1 \right]}. \quad (53)$$

Remark 10.1 (Note on the original paper's gradient expressions). *The gradient expressions given in the original paper (Eqs. (7)–(8) of Tzelepis et al. (2017)) appear to contain transcription errors. Specifically, their Eq. (7) omits the factor y_i in the term multiplying $\boldsymbol{\mu}_i$ (their $\mathbf{x}_i$), and their Eq. (8) omits both the factor y_i and the factor $\frac{1}{2}$ from the bias gradient. Our expressions eqs. (51) and (53) follow from a careful application of the chain rule, as shown above, and can be verified against numerical (finite-difference) gradients.*

10.8 Computational Cost

The key computational cost per training example is:

- $\mathbf{w}^\top \boldsymbol{\mu}_i$: an inner product, $\mathcal{O}(d)$. Same as standard SVM.
- $\mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}$: a quadratic form. $\mathcal{O}(d^2)$ for a general covariance matrix, but $\mathcal{O}(d)$ if $\boldsymbol{\Sigma}_i$ is diagonal (the common case in practice).

Remark 10.2 (Sanity check). *Both $\mathbf{w}^\top \boldsymbol{\mu}_i$ and $\mathbf{w}^\top \boldsymbol{\Sigma}_i \mathbf{w}$ need to be computed only once per example per iteration, since they appear in both the loss and the gradient formulas. So the cost of SVM-GSU with diagonal covariances is essentially the same as the cost of standard SVM.*

11 Convexity of the SVM-GSU Objective

Proposition 11.1. *The SVM-GSU objective $\mathcal{J}_{\text{GSU}}(\mathbf{w}, b)$ given by eq. (31) is strongly convex with respect to $(\mathbf{w}, b)$.*

Proof. We break the objective into parts and use standard preservation-of-convexity results.

Regularizer. The function $(\mathbf{w}, b) \mapsto \frac{\lambda}{2} \|\mathbf{w}\|^2$ is strongly convex in $\mathbf{w}$ (its Hessian is $\lambda \mathbf{I}$, positive definite) and constant (hence convex) in b . Therefore, it is strongly convex in $(\mathbf{w}, b)$ jointly.

Each per-example loss. Consider the integrand at a fixed $\mathbf{x}$:

$$h_i(\mathbf{w}, b; \mathbf{x}) = \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)). \quad (54)$$

The argument inside the max, $1 - y_i(\mathbf{w}^\top \mathbf{x} + b)$, is an affine (hence convex) function of $(\mathbf{w}, b)$. The function $\max(0, \cdot)$ is convex and non-decreasing. The composition of a convex non-decreasing function with a convex function is convex. Hence $h_i(\cdot, \cdot; \mathbf{x})$ is convex in $(\mathbf{w}, b)$ for every fixed $\mathbf{x}$.

The per-example loss is

$$\mathcal{L}_{\text{GSU}_i}(\mathbf{w}, b) = \int_{\mathbb{R}^d} h_i(\mathbf{w}, b; \mathbf{x}) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}. \quad (55)$$

This is an integral of convex functions against a non-negative weight $f_{\mathbf{x}_i}(\mathbf{x})$. Equivalently, for any $\alpha \in [0, 1]$ and any $(\mathbf{w}_1, b_1), (\mathbf{w}_2, b_2)$:

$$\begin{aligned} \mathcal{L}_{\text{GSU}_i}(\alpha(\mathbf{w}_1, b_1) + (1 - \alpha)(\mathbf{w}_2, b_2)) &= \int h_i(\alpha(\mathbf{w}_1, b_1) + (1 - \alpha)(\mathbf{w}_2, b_2); \mathbf{x}) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x} \\ &\leq \int [\alpha h_i(\mathbf{w}_1, b_1; \mathbf{x}) + (1 - \alpha) h_i(\mathbf{w}_2, b_2; \mathbf{x})] f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x} \\ &= \alpha \mathcal{L}_{\text{GSU}_i}(\mathbf{w}_1, b_1) + (1 - \alpha) \mathcal{L}_{\text{GSU}_i}(\mathbf{w}_2, b_2). \end{aligned} \quad (56)$$

So each $\mathcal{L}_{\text{GSU}_i}$ is convex.

Conclusion. A non-negative combination of convex functions is convex, so the average $\frac{1}{n} \sum_i \mathcal{L}_{\text{GSU}_i}$ is convex. Adding the strongly convex regularizer yields a strongly convex objective, which therefore has a unique global minimum. $\square$

12 The SGD Solver (Pegasos-Style)

The SVM-GSU optimization problem is solved using a stochastic gradient descent (SGD) algorithm inspired by the Pegasos algorithm. At each iteration, instead of computing the gradient using the entire training set (which can be expensive for large n), we randomly sample a small subset (“mini-batch”) and compute an approximate gradient.

12.1 Algorithm Description

Algorithm 1 presents the stochastic sub-gradient descent procedure for solving the SVM-GSU optimization problem.

Algorithm 1 Stochastic sub-gradient descent for SVM-GSU (Pegasos-style).

Require: Training set $\mathcal{X}'$, regularization parameter $\lambda > 0$, number of iterations $T \in \mathbb{N}$, mini-batch size $k \in \mathbb{N}$.

Ensure: Learned parameters $\mathbf{w}^{(T+1)}$, $b^{(T+1)}$.

```

1: Initialize  $b^{(1)} \leftarrow 0$  and  $\mathbf{w}^{(1)}$  such that  $\|\mathbf{w}^{(1)}\| \leq 1/\sqrt{\lambda}$ .
2: for  $t = 1, 2, \dots, T$  do
3:   Sample a random mini-batch  $\mathcal{X}_t \subseteq \mathcal{X}'$  with  $|\mathcal{X}_t| = k$ .
4:   Set learning rate  $\eta_t \leftarrow 1/\lambda t$ .
5:   Compute  $\frac{\partial \mathcal{J}_{\text{GSU}}}{\partial \mathbf{w}}$  and  $\frac{\partial \mathcal{J}_{\text{GSU}}}{\partial b}$  using eqs. (51) and (53) on  $\mathcal{X}_t$ .
6:    $\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} - \eta_t \frac{\partial \mathcal{J}_{\text{GSU}}}{\partial \mathbf{w}}$  ▷ Gradient step on  $\mathbf{w}$ 
7:    $\mathbf{w}^{(t+1)} \leftarrow \min\left(1, \frac{1}{\sqrt{\lambda} \|\mathbf{w}^{(t+1)}\|}\right) \mathbf{w}^{(t+1)}$  ▷ Project onto  $\|\mathbf{w}\| \leq 1/\sqrt{\lambda}$ 
8:    $b^{(t+1)} \leftarrow b^{(t)} - \eta_t \frac{\partial \mathcal{J}_{\text{GSU}}}{\partial b}$  ▷ Gradient step on  $b$ 
9: end for
10: return  $\mathbf{w}^{(T+1)}$ ,  $b^{(T+1)}$ 
```

13 Learning in Linear Subspaces

When the dimensionality d is very large, but most of the uncertainty (variance) of each Gaussian lies in a low-dimensional subspace, we can speed up computation by projecting into that subspace. For each training example i , we perform eigendecomposition of Σ_i and keep the top d_i eigenvectors preserving a fraction p of the total variance, giving a projection matrix $\mathbf{P}_i \in \mathbb{R}^{d_i \times d}$. The projected quantities $\mathbf{z}_i = \mathbf{P}_i \boldsymbol{\mu}_i$ and $\Sigma_i^z = \mathbf{P}_i \Sigma_i \mathbf{P}_i^\top$ (which is diagonal) yield the same loss form as eq. (30) in the lower-dimensional space.

14 To Sample or Not to Sample?

One might wonder: why not just draw many samples from each Gaussian and train a standard SVM? A concentration inequality by Cirel'son et al. (2006) shows:

$$P\left(\left|\mathcal{L}_{\text{GSU}_i} - \tilde{\mathcal{L}}_{\text{GSU}_{i,N}}\right| \geq r\right) \leq 2 \exp\left(-\frac{r^2}{2\|\mathbf{w}\|^2}\right), \quad (57)$$

where $\tilde{\mathcal{L}}_{\text{GSU}_{i,N}}$ is the sample-based approximation using N samples. The bound on the right-hand side depends on $\|\mathbf{w}\|^2$. Although the Pegasos algorithm constrains $\|\mathbf{w}\|$ to a ball of radius $1/\sqrt{\lambda}$,

the fundamental difficulty with sampling in high-dimensional spaces is that Gaussian distributions spread their probability mass more thinly as the dimensionality grows. Consequently, far more samples are needed to adequately represent the distribution in higher dimensions. In experiments, approximately 10^3 samples per Gaussian were needed in 2D and 5×10^4 in 3D to match SVM-GSU's hyperplane. In high-dimensional problems (d in the hundreds or thousands), sampling becomes infeasible, making the closed-form approach essential.

Part IV

Extension: SVM with Gaussian Mixture Uncertainty (SVM-GMU)

Roadmap for Part IV

We will proceed in four steps, mirroring the structure of Part III:

1. **Motivate** why Gaussian uncertainty is not always enough (section 15).
2. **Write down** the objective function with a GMM density inside the integral (section 18).
3. **Swap** the sum and integral (linearity of integration), recognizing each term as an SVM-GSU loss we already know how to evaluate.
4. **Substitute** the SVM-GSU closed form to get the fully explicit SVM-GMU objective, then **differentiate** it.

As before, we will display the evolving objective at each stage.

15 Motivation: Why Go Beyond Gaussian Uncertainty?

The SVM-GSU framework assumes that the uncertainty of each training example is described by a single multivariate Gaussian. While this is flexible (it allows anisotropic, example-specific uncertainty), Gaussian distributions have a fundamental limitation: their iso-density contours are always *ellipsoids*. This means:

- The uncertainty is always *unimodal* (one peak).
- The shape is always *ellipsoidal*. Complex shapes like banana-shaped, crescent-shaped, or multi-lobed uncertainty regions cannot be captured.
- The distribution is always *symmetric* around its mean. Skewed uncertainty is not representable.

16 Gaussian Mixture Models (GMMs) as Universal Approximators

The solution is to model the uncertainty as a *Gaussian Mixture Model* (GMM). A GMM is a weighted sum of several Gaussian distributions:

$$f_{\mathbf{x}_i}(\mathbf{x}) = \sum_{m=1}^{M_i} \pi_i^{(m)} \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)}), \quad (58)$$

where M_i is the number of components, $\pi_i^{(m)} \geq 0$ are mixing weights with $\sum_m \pi_i^{(m)} = 1$, and each component has mean $\boldsymbol{\mu}_i^{(m)} \in \mathbb{R}^d$ and covariance $\boldsymbol{\Sigma}_i^{(m)} \in \mathbb{S}_{++}^d$.

GMMs are *universal density approximators*: given enough components, a GMM can approximate any continuous probability distribution to arbitrary accuracy.

17 The New Training Set for SVM-GMU

The new training set considering Gaussian mixture uncertainty is then:

$$\mathcal{X}'' = \left\{ \left(\left\{ \pi_i^{(m)}, \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)} \right\}_{m=1}^{M_i}, y_i \right) : i = 1, \dots, n \right\}. \quad (59)$$

18 Deriving the SVM-GMU Objective Function

18.1 Stage 1: The Objective with the GMM Integral

Just as in SVM-GSU, we replace the hinge loss with its expected value. The objective has the same structure as eq. (15), but with the GMM density $f_{\mathbf{x}_i}$ from eq. (58):

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \underbrace{\frac{1}{n} \sum_{i=1}^n \int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)}. \quad (60)$$

SVM-GMU Objective - Stage 1: Integral Form (eq. (60))

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \underbrace{\frac{1}{n} \sum_{i=1}^n \int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)}.$$

Same structure as SVM-GSU (eq. (15)), but $f_{\mathbf{x}_i}$ is now a GMM, not a single Gaussian. The integral $\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)$ is not yet tractable.

18.2 Stage 2: Substituting the GMM Density and Swapping Sum and Integral

In eq. (60), we now substitute $f_{\mathbf{x}_i}(\mathbf{x})$ with the GMM density from eq. (58):

$$\begin{aligned} \mathcal{J}_{\text{GMU}}(\mathbf{w}, b) &= \frac{\lambda}{2} \|\mathbf{w}\|^2 \\ &+ \underbrace{\frac{1}{n} \sum_{i=1}^n \int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) \left[\sum_{m=1}^{M_i} \pi_i^{(m)} \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)}) \right] d\mathbf{x}}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)}. \end{aligned} \quad (61)$$

Justification for swapping sum and integral. We use the linearity of the integral. The integrand is a finite sum of non-negative integrable terms (the hinge loss is non-negative, each Gaussian PDF is non-negative, and each $\pi_i^{(m)} \geq 0$), so Tonelli's theorem justifies exchanging the

finite sum and the integral:

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \underbrace{\frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)}) d\mathbf{x}}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)} \quad (62)$$

$\mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b) = \text{SVM-GSU loss for component } m$

Each integral $\mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b)$ is *exactly* the SVM-GSU loss (eq. (30)) for a Gaussian with mean $\boldsymbol{\mu}_i^{(m)}$ and covariance $\boldsymbol{\Sigma}_i^{(m)}$. We already know its closed form.

SVM-GMU Objective - Stage 2: Mixture Form (eq. (62))

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \underbrace{\frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)}) d\mathbf{x}}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)},$$

$\mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b) = \text{SVM-GSU loss for component } m$

Each $\mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b)$ is still an integral, but one we already evaluated in closed form in section 9.

18.3 Stage 3: The Fully Explicit SVM-GMU Objective

Applying the closed-form loss eq. (30) to each component, with:

$$d_{\boldsymbol{\mu}_i}^{(m)} = 1 - y_i(\mathbf{w}^\top \boldsymbol{\mu}_i^{(m)} + b), \quad (63)$$

$$d_{\boldsymbol{\Sigma}_i}^{(m)} = \sqrt{2 \mathbf{w}^\top \boldsymbol{\Sigma}_i^{(m)} \mathbf{w}}, \quad (64)$$

we get the component loss:

$$\mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b) = \frac{d_{\boldsymbol{\mu}_i}^{(m)}}{2} \left[\text{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right] + \frac{d_{\boldsymbol{\Sigma}_i}^{(m)}}{2\sqrt{\pi}} \exp \left(- \frac{(d_{\boldsymbol{\mu}_i}^{(m)})^2}{(d_{\boldsymbol{\Sigma}_i}^{(m)})^2} \right). \quad (65)$$

The total loss for the i -th example is the weighted sum:

$$\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b) = \sum_{m=1}^{M_i} \pi_i^{(m)} \mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b). \quad (66)$$

And the **full SVM-GMU objective** is:

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \underbrace{\frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \left[\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{2} \left(\text{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right) + \frac{d_{\boldsymbol{\Sigma}_i}^{(m)}}{2\sqrt{\pi}} \exp \left(- \frac{(d_{\boldsymbol{\mu}_i}^{(m)})^2}{(d_{\boldsymbol{\Sigma}_i}^{(m)})^2} \right) \right]}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)}. \quad (67)$$

$\mathcal{L}_{\text{GSU}_i^{(m)}}(\mathbf{w}, b)$

$\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)$

SVM-GMU Objective - Stage 3: Fully Explicit Closed Form (eq. (67))

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \underbrace{\frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \left[\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{2} \left(\operatorname{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right) + \frac{d_{\boldsymbol{\Sigma}_i}^{(m)}}{2\sqrt{\pi}} \exp \left(-\frac{(d_{\boldsymbol{\mu}_i}^{(m)})^2}{(d_{\boldsymbol{\Sigma}_i}^{(m)})^2} \right) \right]}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)}$$

The only unknowns are still $\mathbf{w}$ and b . Compared to SVM-GSU (eq. (31)), the only structural difference is the extra sum over components m with weights $\pi_i^{(m)}$. No integrals remain. This is what we minimize.

Remark 18.1 (Sanity check). Setting $M_i = 1$ and $\pi_i^{(1)} = 1$ for all i collapses the inner sum to a single term, and eq. (67) reduces exactly to the SVM-GSU objective eq. (31). Further setting $\boldsymbol{\Sigma}_i^{(1)} \rightarrow \mathbf{0}$ recovers the standard SVM. The nesting is: Standard SVM $\subset$ SVM-GSU $\subset$ SVM-GMU.

19 Gradients of the SVM-GMU Objective

We differentiate the closed-form objective eq. (67). Since it has the form

$$\frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_i \sum_m \pi_i^{(m)} \mathcal{L}_{\text{GSU}_i}^{(m)}(\mathbf{w}, b),$$

and differentiation is linear, each gradient is the regularizer's contribution plus a weighted sum of the per-component gradients. Each per-component gradient is exactly the SVM-GSU gradient from eqs. (51) and (53), applied to the component parameters $\boldsymbol{\mu}_i^{(m)}$ and $\boldsymbol{\Sigma}_i^{(m)}$.

19.1 Derivation of the Gradient with Respect to $\mathbf{w}$

Starting from the gradient of a single SVM-GSU component loss (derived in section 10):

$$\frac{\partial \mathcal{L}_{\text{GSU}_i}^{(m)}}{\partial \mathbf{w}} = \frac{\exp \left(-\frac{(d_{\boldsymbol{\mu}_i}^{(m)})^2}{(d_{\boldsymbol{\Sigma}_i}^{(m)})^2} \right)}{\sqrt{\pi} d_{\boldsymbol{\Sigma}_i}^{(m)}} \boldsymbol{\Sigma}_i^{(m)} \mathbf{w} - \frac{y_i}{2} \left[\operatorname{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right] \boldsymbol{\mu}_i^{(m)}. \quad (68)$$

By linearity of differentiation applied to eq. (67):

$$\frac{\partial \mathcal{J}_{\text{GMU}}}{\partial \mathbf{w}} = \lambda \mathbf{w} + \frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \frac{\partial \mathcal{L}_{\text{GSU}_i}^{(m)}}{\partial \mathbf{w}}. \quad (69)$$

Substituting:

$$\frac{\partial \mathcal{J}_{\text{GMU}}}{\partial \mathbf{w}} = \lambda \mathbf{w} + \frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \left[\frac{\exp \left(-\frac{(d_{\boldsymbol{\mu}_i}^{(m)})^2}{(d_{\boldsymbol{\Sigma}_i}^{(m)})^2} \right)}{\sqrt{\pi} d_{\boldsymbol{\Sigma}_i}^{(m)}} \boldsymbol{\Sigma}_i^{(m)} \mathbf{w} - \frac{y_i}{2} \left(\operatorname{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right) \boldsymbol{\mu}_i^{(m)} \right]. \quad (70)$$

19.2 Derivation of the Gradient with Respect to b

Similarly, the per-component bias gradient is

$$\frac{\partial \mathcal{L}_{\text{GSU}_i}^{(m)}}{\partial b} = -\frac{y_i}{2} \left[\text{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right]. \quad (71)$$

Summing:

$$\boxed{\frac{\partial \mathcal{J}_{\text{GMU}}}{\partial b} = -\frac{1}{n} \sum_{i=1}^n \sum_{m=1}^{M_i} \pi_i^{(m)} \frac{y_i}{2} \left[\text{erf} \left(\frac{d_{\boldsymbol{\mu}_i}^{(m)}}{d_{\boldsymbol{\Sigma}_i}^{(m)}} \right) + 1 \right]}. \quad (72)$$

Remark 19.1 (Sanity check). Compare eq. (70) with the SVM-GSU gradient eq. (51). The structure is identical; each term inside the brackets has the same form, but now there is an additional weighted sum over components m . If $M_i = 1$ and $\pi_i^{(1)} = 1$, the extra sum collapses to a single term and we recover the SVM-GSU gradient exactly.

20 Convexity of the SVM-GMU Objective

Proposition 20.1. $\mathcal{J}_{\text{GMU}}(\mathbf{w}, b)$ is strongly convex with respect to $(\mathbf{w}, b)$.

Proof. By theorem 11.1 and the reasoning in its proof, each per-component loss $\mathcal{L}_{\text{GSU}_i}^{(m)}(\mathbf{w}, b)$ is convex in $(\mathbf{w}, b)$. Since $\pi_i^{(m)} \geq 0$, the mixture loss

$$\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b) = \sum_{m=1}^{M_i} \pi_i^{(m)} \mathcal{L}_{\text{GSU}_i}^{(m)}(\mathbf{w}, b) \quad (73)$$

is a non-negative weighted sum of convex functions, hence convex. The average over training examples is therefore also convex. Adding the strongly convex regularizer $\frac{\lambda}{2} \|\mathbf{w}\|^2$ yields a strongly convex objective with a unique global minimum. $\square$

21 The SGD Algorithm for SVM-GMU

The Pegasos-style SGD extends directly from algorithm 1. The only structural change is that gradient computation now sums over M_i components per example using eqs. (70) and (72). Per-example cost is $\mathcal{O}(M_i \cdot d)$ with diagonal covariances, compared to $\mathcal{O}(d)$ for SVM-GSU. The full algorithm is given in algorithm 2.

Algorithm 2 Stochastic sub-gradient descent for SVM-GMU.

Require: Training set $\mathcal{X}''$ of GMM-annotated examples, regularization parameter $\lambda > 0$, number of iterations $T \in \mathbb{N}$, mini-batch size $k \in \mathbb{N}$.

Ensure: Learned parameters $\mathbf{w}^{(T+1)}$, $b^{(T+1)}$.

```
1: Initialize  $b^{(1)} \leftarrow 0$  and  $\mathbf{w}^{(1)}$  such that  $\|\mathbf{w}^{(1)}\| \leq 1/\sqrt{\lambda}$ .
2: for  $t = 1, 2, \dots, T$  do
3:   Sample a random mini-batch  $\mathcal{X}_t \subseteq \mathcal{X}''$  with  $|\mathcal{X}_t| = k$ .
4:   Set learning rate  $\eta_t \leftarrow 1/\lambda t$ .
5:   Initialize gradient accumulators  $\mathbf{g}_w \leftarrow \mathbf{0}$ ,  $g_b \leftarrow 0$ .
6:   for each example  $i$  in  $\mathcal{X}_t$  do
7:     for each component  $m = 1, \dots, M_i$  do
8:       Compute  $d_{\mu_i}^{(m)}$  and  $d_{\Sigma_i}^{(m)}$  using eqs. (63) and (64).
9:       Accumulate  $\mathbf{g}_w \leftarrow \mathbf{g}_w + \pi_i^{(m)} \cdot$  (per-component contribution from eq. (70)).
10:      Accumulate  $g_b \leftarrow g_b + \pi_i^{(m)} \cdot$  (per-component contribution from eq. (72)).
11:    end for
12:  end for
13:  Add regularizer gradient:  $\mathbf{g}_w \leftarrow \mathbf{g}_w + \lambda \mathbf{w}^{(t)}$ .
14:   $\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} - \eta_t \mathbf{g}_w$  ▷ Gradient step on  $\mathbf{w}$ 
15:   $\mathbf{w}^{(t+1)} \leftarrow \min\left(1, \frac{1}{\sqrt{\lambda} \|\mathbf{w}^{(t+1)}\|}\right) \mathbf{w}^{(t+1)}$  ▷ Project onto  $\|\mathbf{w}\| \leq 1/\sqrt{\lambda}$ 
16:   $b^{(t+1)} \leftarrow b^{(t)} - \eta_t g_b$  ▷ Gradient step on  $b$ 
17: end for
18: return  $\mathbf{w}^{(T+1)}$ ,  $b^{(T+1)}$ 
```

22 Limiting and Special Cases

- $M_i = 1$ for all i : recovers SVM-GSU.
- $M_i = 1$ and $\Sigma_i \rightarrow \mathbf{0}$: recovers standard SVM.
- Shared means ($\mu_i^{(m)} = \mu_i$ for all m): models heavier-than-Gaussian tails via a scale mixture.
- Isotropic components ($\Sigma_i^{(m)} = (\sigma_i^{(m)})^2 \mathbf{I}$): $d_{\Sigma_i}^{(m)} = \sqrt{2} \sigma_i^{(m)} \|\mathbf{w}\|$.

23 Extension to Linear Subspace Learning

Two strategies:

1. per-component subspaces via eigendecomposition of each $\Sigma_i^{(m)}$, or
2. a shared subspace per example.

For strategy (2), we can use the weighted mean covariance $\bar{\Sigma}_i = \sum_m \pi_i^{(m)} \Sigma_i^{(m)}$, or the full mixture covariance given by the *law of total covariance*:

$$\text{Cov}(\mathbf{x}_i) = \bar{\Sigma}_i + \sum_{m=1}^{M_i} \pi_i^{(m)} \left(\mu_i^{(m)} - \bar{\mu}_i \right) \left(\mu_i^{(m)} - \bar{\mu}_i \right)^\top, \quad \bar{\mu}_i = \sum_m \pi_i^{(m)} \mu_i^{(m)}. \quad (74)$$

The first term captures the average within-component variance; the second captures the variance of the means. This is the correct covariance of a random vector drawn from the GMM.

24 Summary Table

Table 1: Comparison of the standard SVM, SVM-GSU, and the proposed SVM-GMU.

Property	Standard SVM	SVM-GSU	SVM-GMU
Uncertainty model	None (point)	Gaussian	Gaussian mixture
Uncertainty shape	—	Ellipsoidal	Arbitrary
Multimodal	—	No	Yes
Components per example	—	1	$M_i \geq 1$
Closed-form loss	Yes	Yes	Yes
Closed-form gradients	Yes	Yes	Yes
Convex objective	Yes	Yes	Yes
SGD applicable	Yes	Yes	Yes
Per-example cost (diagonal Σ)	$\mathcal{O}(d)$	$\mathcal{O}(d)$	$\mathcal{O}(M_i \cdot d)$
Degenerates to standard SVM	N/A	Yes ($\Sigma_i \rightarrow \mathbf{0}$)	Yes ($M_i = 1, \Sigma_i \rightarrow \mathbf{0}$)

Part V

Experiments and Empirical Results

This part presents four experiments carried out with the author’s Python implementation of SVM-GMU. The first (section 25) introduces a small two-dimensional dataset with deliberately non-Gaussian per-sample uncertainty and shows how the learned SVM-GMU boundary differs from the standard SVM boundary. The second (section 27) empirically verifies the claim of section 14: a standard SVM trained on samples drawn from the per-sample GMMs converges, as the number of samples grows, to the SVM-GMU boundary obtained in closed form on the original data. The third (section 28) isolates the contribution of the mixture structure itself by comparing SVM-GMU against SVM-GSU fitted on a moment-matched single-Gaussian approximation of the same GMMs. The fourth (section 29) takes the realistic next step: rather than assuming the per-sample GMMs are given, it draws samples from each true GMM, re-estimates a GMM per sample by Expectation Maximization with the component count chosen by BIC, and shows that the SVM-GMU boundary fit on the estimated mixtures converges to the closed-form reference as the sample size grows.

All four experiments share a common dataset, which we describe in detail in section 25 and then reuse throughout.

25 The close_separable Dataset

25.1 Overview

We construct a dataset with $n = 6$ training examples in $d = 2$ dimensions. Three examples belong to class +1 and three to class −1. Each example’s uncertainty is modeled as a Gaussian mixture

with $M_i \in \{5, 6\}$ components, chosen so that the iso-density contours of the mixtures trace out *banana*- and *crescent*-shaped curves. The two class regions are placed close enough to each other that the orientation of the uncertainty has a visible effect on the learned decision boundary, while still being linearly separable at the level of the observed points themselves. This makes it a useful sandbox for studying how uncertainty-aware training differs from the standard SVM.

25.2 Observed Feature Vectors and Class Labels

The six observed feature vectors (the overall means, in the notation of section 17) are stored as the rows of the matrix $\mathbf{X} \in \mathbb{R}^{6 \times 2}$, and the corresponding class labels form the vector $\mathbf{y} \in \{+1, -1\}^6$:

$$\mathbf{X} = \begin{bmatrix} -1.8 & 0.0 \\ -2.0 & 1.5 \\ -1.0 & 2.5 \\ 1.5 & -1.0 \\ 1.0 & 1.0 \\ 2.0 & 2.0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} +1 \\ +1 \\ +1 \\ -1 \\ -1 \\ -1 \end{bmatrix}. \quad (75)$$

The first three rows of $\mathbf{X}$ form the positive class, the last three form the negative class.

25.3 Per-Sample GMM Uncertainties

Each observed point is accompanied by a GMM with between 5 and 6 components. The i -th training example is therefore a tuple

$$\left(\left\{ \pi_i^{(m)}, \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)} \right\}_{m=1}^{M_i}, y_i \right), \quad (76)$$

matching the SVM-GMU training-set definition in eq. (59). The component means are placed along curved arcs, and each component covariance is elongated roughly along the local tangent of the arc so that the mixture of ellipses traces out a banana or a crescent rather than a single fat ellipse.

We now list the six GMMs explicitly.

Sample $i = 1$ ($y_1 = +1$). Five components with mixing weights and means

$$\boldsymbol{\pi}_1 = \begin{bmatrix} 0.10 \\ 0.20 \\ 0.40 \\ 0.20 \\ 0.10 \end{bmatrix}, \quad \left\{ \boldsymbol{\mu}_1^{(m)} \right\}_{m=1}^5 = \left\{ \begin{bmatrix} -4.2 \\ -0.3 \end{bmatrix}, \begin{bmatrix} -3.8 \\ 0.3 \end{bmatrix}, \begin{bmatrix} -3.0 \\ 0.5 \end{bmatrix}, \begin{bmatrix} -2.2 \\ 0.3 \end{bmatrix}, \begin{bmatrix} -1.8 \\ -0.3 \end{bmatrix} \right\}, \quad (77)$$

and full covariance matrices

$$\begin{aligned} \boldsymbol{\Sigma}_1^{(1)} &= \begin{bmatrix} 0.08 & -0.05 \\ -0.05 & 0.15 \end{bmatrix}, & \boldsymbol{\Sigma}_1^{(2)} &= \begin{bmatrix} 0.10 & -0.03 \\ -0.03 & 0.12 \end{bmatrix}, & \boldsymbol{\Sigma}_1^{(3)} &= \begin{bmatrix} 0.18 & 0.00 \\ 0.00 & 0.06 \end{bmatrix}, \\ \boldsymbol{\Sigma}_1^{(4)} &= \begin{bmatrix} 0.10 & 0.03 \\ 0.03 & 0.12 \end{bmatrix}, & \boldsymbol{\Sigma}_1^{(5)} &= \begin{bmatrix} 0.08 & 0.05 \\ 0.05 & 0.15 \end{bmatrix}. \end{aligned} \quad (78)$$

Sample $i = 2$ ($y_2 = +1$). Six components with mixing weights and means

$$\boldsymbol{\pi}_2 = \begin{bmatrix} 0.08 \\ 0.15 \\ 0.22 \\ 0.30 \\ 0.15 \\ 0.10 \end{bmatrix}, \quad \left\{ \boldsymbol{\mu}_2^{(m)} \right\}_{m=1}^6 = \left\{ \begin{bmatrix} -2.2 \\ 0.6 \end{bmatrix}, \begin{bmatrix} -1.8 \\ 1.1 \end{bmatrix}, \begin{bmatrix} -1.4 \\ 1.5 \end{bmatrix}, \begin{bmatrix} -0.8 \\ 1.7 \end{bmatrix}, \begin{bmatrix} -0.4 \\ 1.5 \end{bmatrix}, \begin{bmatrix} -0.1 \\ 1.1 \end{bmatrix} \right\}, \quad (79)$$

and full covariance matrices

$$\begin{aligned} \boldsymbol{\Sigma}_2^{(1)} &= \begin{bmatrix} 0.07 & 0.04 \\ 0.04 & 0.13 \end{bmatrix}, & \boldsymbol{\Sigma}_2^{(2)} &= \begin{bmatrix} 0.09 & 0.03 \\ 0.03 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_2^{(3)} &= \begin{bmatrix} 0.14 & 0.02 \\ 0.02 & 0.07 \end{bmatrix}, \\ \boldsymbol{\Sigma}_2^{(4)} &= \begin{bmatrix} 0.18 & 0.00 \\ 0.00 & 0.05 \end{bmatrix}, & \boldsymbol{\Sigma}_2^{(5)} &= \begin{bmatrix} 0.12 & -0.02 \\ -0.02 & 0.07 \end{bmatrix}, & \boldsymbol{\Sigma}_2^{(6)} &= \begin{bmatrix} 0.08 & -0.04 \\ -0.04 & 0.12 \end{bmatrix}. \end{aligned} \quad (80)$$

Sample $i = 3$ ($y_3 = +1$). Five components with mixing weights and means

$$\boldsymbol{\pi}_3 = \begin{bmatrix} 0.10 \\ 0.20 \\ 0.40 \\ 0.20 \\ 0.10 \end{bmatrix}, \quad \left\{ \boldsymbol{\mu}_3^{(m)} \right\}_{m=1}^5 = \left\{ \begin{bmatrix} -0.6 \\ 2.2 \end{bmatrix}, \begin{bmatrix} -0.2 \\ 2.7 \end{bmatrix}, \begin{bmatrix} 0.3 \\ 3.0 \end{bmatrix}, \begin{bmatrix} 0.8 \\ 3.4 \end{bmatrix}, \begin{bmatrix} 1.1 \\ 3.8 \end{bmatrix} \right\}, \quad (81)$$

and full covariance matrices

$$\begin{aligned} \boldsymbol{\Sigma}_3^{(1)} &= \begin{bmatrix} 0.14 & -0.06 \\ -0.06 & 0.08 \end{bmatrix}, & \boldsymbol{\Sigma}_3^{(2)} &= \begin{bmatrix} 0.12 & -0.07 \\ -0.07 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_3^{(3)} &= \begin{bmatrix} 0.10 & -0.05 \\ -0.05 & 0.14 \end{bmatrix}, \\ \boldsymbol{\Sigma}_3^{(4)} &= \begin{bmatrix} 0.08 & 0.03 \\ 0.03 & 0.16 \end{bmatrix}, & \boldsymbol{\Sigma}_3^{(5)} &= \begin{bmatrix} 0.10 & 0.06 \\ 0.06 & 0.15 \end{bmatrix}. \end{aligned} \quad (82)$$

Sample $i = 4$ ($y_4 = -1$). Six components with mixing weights and means

$$\boldsymbol{\pi}_4 = \begin{bmatrix} 0.08 \\ 0.17 \\ 0.25 \\ 0.25 \\ 0.17 \\ 0.08 \end{bmatrix}, \quad \left\{ \boldsymbol{\mu}_4^{(m)} \right\}_{m=1}^6 = \left\{ \begin{bmatrix} -0.9 \\ -0.3 \end{bmatrix}, \begin{bmatrix} -0.5 \\ -0.7 \end{bmatrix}, \begin{bmatrix} -0.1 \\ -1.0 \end{bmatrix}, \begin{bmatrix} 0.4 \\ -1.2 \end{bmatrix}, \begin{bmatrix} 0.8 \\ -1.1 \end{bmatrix}, \begin{bmatrix} 1.1 \\ -0.8 \end{bmatrix} \right\}, \quad (83)$$

and full covariance matrices

$$\begin{aligned} \boldsymbol{\Sigma}_4^{(1)} &= \begin{bmatrix} 0.07 & 0.04 \\ 0.04 & 0.12 \end{bmatrix}, & \boldsymbol{\Sigma}_4^{(2)} &= \begin{bmatrix} 0.09 & 0.03 \\ 0.03 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_4^{(3)} &= \begin{bmatrix} 0.15 & 0.00 \\ 0.00 & 0.07 \end{bmatrix}, \\ \boldsymbol{\Sigma}_4^{(4)} &= \begin{bmatrix} 0.15 & 0.00 \\ 0.00 & 0.07 \end{bmatrix}, & \boldsymbol{\Sigma}_4^{(5)} &= \begin{bmatrix} 0.09 & -0.03 \\ -0.03 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_4^{(6)} &= \begin{bmatrix} 0.12 & -0.05 \\ -0.05 & 0.09 \end{bmatrix}. \end{aligned} \quad (84)$$

Sample $i = 5$ ($y_5 = -1$). Five components with mixing weights and means

$$\boldsymbol{\pi}_5 = \begin{bmatrix} 0.10 \\ 0.20 \\ 0.40 \\ 0.20 \\ 0.10 \end{bmatrix}, \quad \left\{ \boldsymbol{\mu}_5^{(m)} \right\}_{m=1}^5 = \left\{ \begin{bmatrix} 1.3 \\ 0.7 \end{bmatrix}, \begin{bmatrix} 1.7 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 2.3 \\ 0.1 \end{bmatrix}, \begin{bmatrix} 2.9 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 3.3 \\ 0.7 \end{bmatrix} \right\}, \quad (85)$$

and full covariance matrices

$$\begin{aligned} \boldsymbol{\Sigma}_5^{(1)} &= \begin{bmatrix} 0.09 & 0.05 \\ 0.05 & 0.14 \end{bmatrix}, & \boldsymbol{\Sigma}_5^{(2)} &= \begin{bmatrix} 0.12 & 0.03 \\ 0.03 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_5^{(3)} &= \begin{bmatrix} 0.16 & 0.00 \\ 0.00 & 0.07 \end{bmatrix}, \\ \boldsymbol{\Sigma}_5^{(4)} &= \begin{bmatrix} 0.12 & -0.03 \\ -0.03 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_5^{(5)} &= \begin{bmatrix} 0.09 & -0.05 \\ -0.05 & 0.14 \end{bmatrix}. \end{aligned} \quad (86)$$

Sample $i = 6$ ($y_6 = -1$). Six components with mixing weights and means

$$\boldsymbol{\pi}_6 = \begin{bmatrix} 0.08 \\ 0.17 \\ 0.25 \\ 0.25 \\ 0.17 \\ 0.08 \end{bmatrix}, \quad \left\{ \boldsymbol{\mu}_6^{(m)} \right\}_{m=1}^6 = \left\{ \begin{bmatrix} 2.4 \\ 2.3 \end{bmatrix}, \begin{bmatrix} 2.8 \\ 2.0 \end{bmatrix}, \begin{bmatrix} 3.3 \\ 2.0 \end{bmatrix}, \begin{bmatrix} 3.8 \\ 2.1 \end{bmatrix}, \begin{bmatrix} 4.2 \\ 2.5 \end{bmatrix}, \begin{bmatrix} 4.4 \\ 3.0 \end{bmatrix} \right\}, \quad (87)$$

and full covariance matrices

$$\begin{aligned} \boldsymbol{\Sigma}_6^{(1)} &= \begin{bmatrix} 0.08 & 0.05 \\ 0.05 & 0.11 \end{bmatrix}, & \boldsymbol{\Sigma}_6^{(2)} &= \begin{bmatrix} 0.12 & 0.03 \\ 0.03 & 0.08 \end{bmatrix}, & \boldsymbol{\Sigma}_6^{(3)} &= \begin{bmatrix} 0.15 & 0.00 \\ 0.00 & 0.06 \end{bmatrix}, \\ \boldsymbol{\Sigma}_6^{(4)} &= \begin{bmatrix} 0.13 & -0.02 \\ -0.02 & 0.07 \end{bmatrix}, & \boldsymbol{\Sigma}_6^{(5)} &= \begin{bmatrix} 0.09 & -0.04 \\ -0.04 & 0.10 \end{bmatrix}, & \boldsymbol{\Sigma}_6^{(6)} &= \begin{bmatrix} 0.07 & -0.05 \\ -0.05 & 0.12 \end{bmatrix}. \end{aligned} \quad (88)$$

25.4 Visualizing the GMM Uncertainty

Before fitting any classifier, we can get a sense of the dataset by plotting the GMM density of each sample. Figure 1 shows the 3σ iso-density contours of each of the six mixtures, colored by class (blue for +1, red for -1). The markings (blue dots and red boxes) mark the observed feature vectors (the rows of $\mathbf{X}$). The banana and crescent shapes are clearly visible and, crucially, extend toward the decision region between the two classes, which is what makes them interesting from a classification point of view: a Gaussian could not faithfully represent where probability mass actually lies.

26 Experiment 1: Fitting SVM-GMU on the `close_separable` Dataset

26.1 Training Setup

We train SVM-GMU on the dataset of section 25 using the Pegasos-style SGD described in algorithm 2. The hyperparameters are:

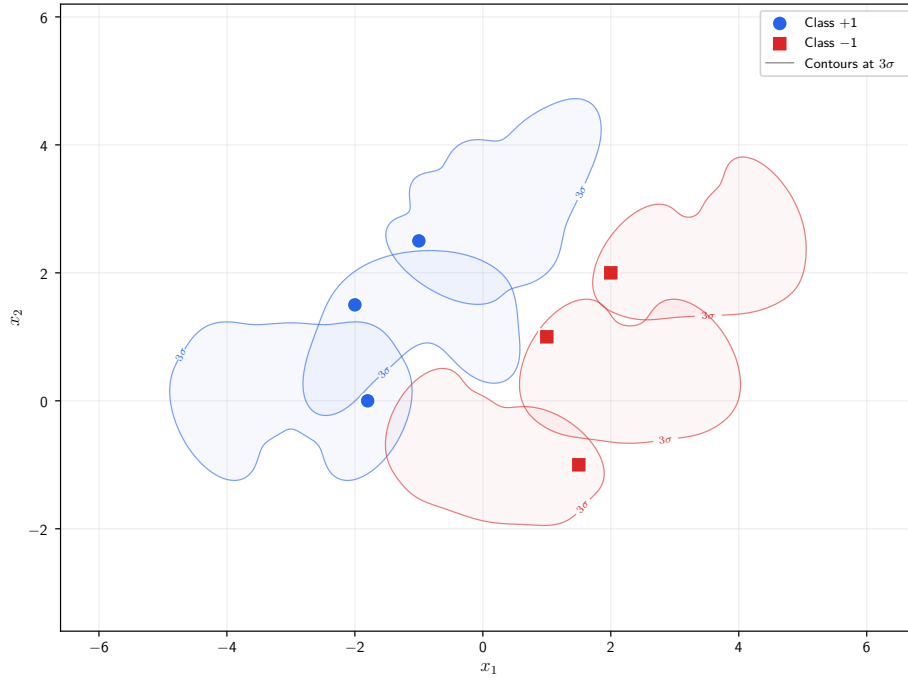


Figure 1: The six per-sample GMM uncertainty distributions of the `close_separable` dataset, visualized as 3σ iso-density contours. Blue contours correspond to class +1, red contours to class -1. Blue dots and red boxes mark the observed feature vectors (the rows of $\mathbf{X}$ in eq. (75)). The banana and crescent shapes cannot be represented by single Gaussians.

- Regularization parameter $\lambda = 10^{-2}$.
- Number of iterations $T = 1000$.
- Mini-batch size $k = 6$, equal to n (i.e., full-batch SGD, since n is small).
- Random seed 42 for reproducibility.

For comparison, we also train a standard linear SVM on the same observed points by invoking the same estimator with the uncertainty description omitted. As shown in section 22, this reduces the objective exactly to the standard hinge-loss SVM objective eq. (9). The standard SVM is trained with $T = 5000$ iterations (the standard hinge loss has flat regions on the correct side of the margin, so convergence of SGD can take slightly longer).

26.2 Learned Parameters and Convergence

The SGD trajectory of SVM-GMU converges rapidly: the objective value drops from roughly 0.0508 at iteration 50 to 0.0472 at iteration 250 and stays essentially flat afterward (the value at iteration 1000 is 0.047228). The learned parameters are

$$\mathbf{w}_{\text{GMU}} = \begin{bmatrix} -1.2527 \\ 1.2589 \end{bmatrix}, \quad b_{\text{GMU}} = -0.9470. \quad (89)$$

The standard SVM converges to a different objective-value plateau (around 0.00366, reflecting the much smaller regularized hinge loss attained by a classifier that is free to ignore the heavy tails of the GMMs), with learned parameters

$$\mathbf{w}_{\text{SVM}} = \begin{bmatrix} -0.8106 \\ 0.2575 \end{bmatrix}, \quad b_{\text{SVM}} = -0.4547. \quad (90)$$

Both models classify all six observed points correctly. Their decision scores $\mathbf{w}^\top \mathbf{x}_i + b$ on the training points are listed in table 2; the sign matches the class label in every row, but the *magnitudes* differ substantially because SVM-GMU is responding to the *entire* GMM density, not just the mean.

Table 2: Decision scores $\mathbf{w}^\top \mathbf{x}_i + b$ of SVM-GMU and the standard SVM on the six observed points of the `close_separable` dataset. Both classifiers predict the correct class (sign) on every point, but their magnitudes differ because SVM-GMU is shaped by the full GMM densities.

i	y_i	SVM-GMU score	Standard SVM score
1	+1	1.3078	1.0044
2	+1	3.4466	1.5528
3	+1	3.4528	0.9997
4	-1	-4.0849	-1.9281
5	-1	-0.9408	-1.0078
6	-1	-0.9346	-1.5609

26.3 Visualizing the SVM-GMU Boundary

Figure 2 overlays the learned SVM-GMU decision boundary $\mathbf{w}_{\text{GMU}}^\top \mathbf{x} + b_{\text{GMU}} = 0$ (solid black) and its margin lines $\mathbf{w}_{\text{GMU}}^\top \mathbf{x} + b_{\text{GMU}} = \pm 1$ (dashed) on top of the GMM contours of fig. 1. The classifier

has rotated its normal to be roughly perpendicular to the arc connecting the two class clouds, which is exactly what we expect given how the uncertainty is oriented in this region.

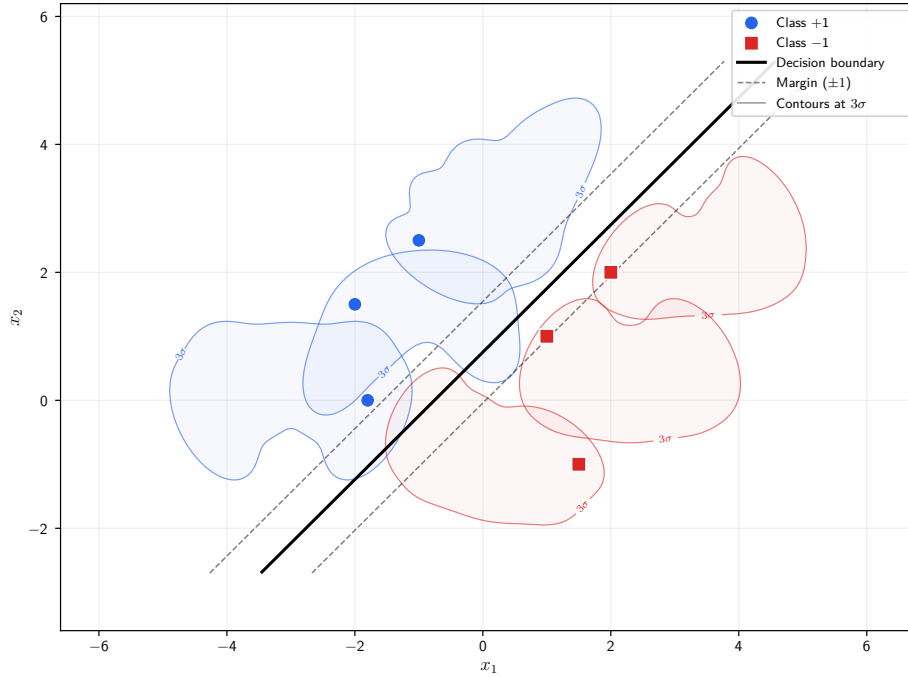


Figure 2: SVM-GMU decision boundary (solid black line) and margin lines (dashed) overlaid on the 3σ GMM contours of the `close_separable` dataset. Learned parameters: $\mathbf{w}_{\text{GMU}} = [-1.2527 \ 1.2589]^\top$, $b_{\text{GMU}} = -0.9470$.

26.4 Comparison with the Standard SVM

Figure 3 shows both decision boundaries on the same plot: SVM-GMU in solid black and the standard SVM in dashed orange. Even though the two classifiers agree on the *labels* of all six observed points, their boundaries are visibly different in orientation and position. The standard SVM responds only to the six observed $\mathbf{x}_i$'s and picks the maximum-margin line that separates those six points. The SVM-GMU boundary, in contrast, is shaped by the full GMM densities: where a GMM extends heavily toward the opposite class (as several of the banana/crescent arcs do), the SVM-GMU pushes the boundary away from the *mass* of that extension, not from the single point that nominally “is” the sample. Concretely, SVM-GMU rotates its normal roughly 27.5° away from the standard SVM normal and places the boundary farther to the right, responding to the rightward-leaning arcs of the positive class and the downward/leftward-leaning arcs of the negative class. This is precisely what an uncertainty-aware classifier should do; the comparison makes the effect visible rather than abstract.

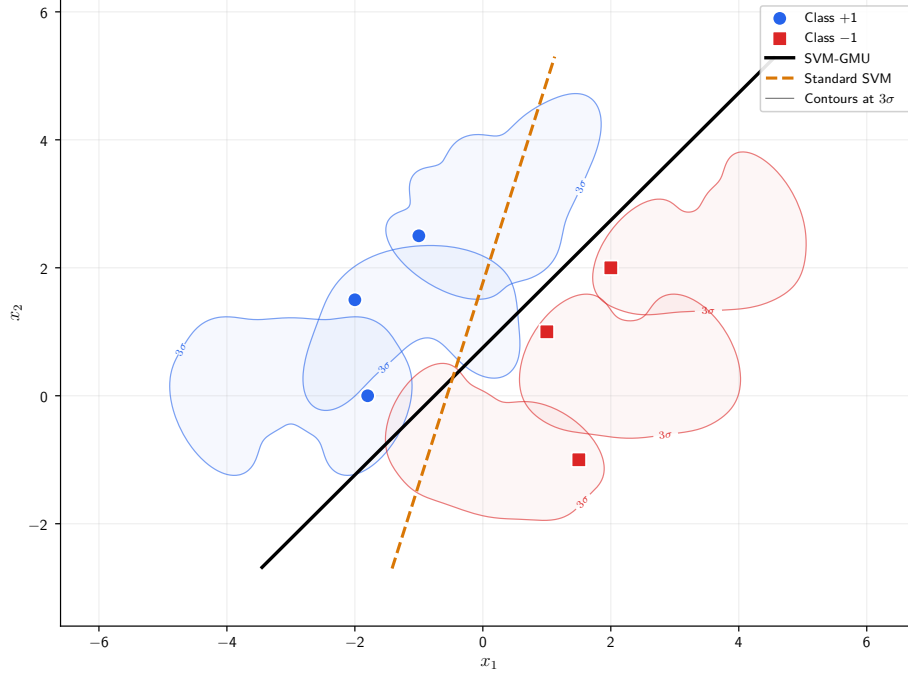


Figure 3: Side-by-side comparison of the SVM-GMU boundary (solid black) and the standard linear SVM boundary (dashed orange) on the `close_separable` dataset, overlaid on the 3σ GMM contours. Both classifiers are trained with the same $\lambda = 10^{-2}$ and random seed. The learned parameters are given in eqs. (89) and (90).

27 Experiment 2: Monte Carlo Convergence of Standard SVM to SVM-GMU

27.1 Motivation

Recall that the SVM-GMU objective eq. (60) is a regularized expected hinge loss,

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_{i=1}^n \underbrace{\int_{\mathbb{R}^d} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b)) f_{\mathbf{x}_i}(\mathbf{x}) \, d\mathbf{x}}_{\mathcal{L}_{\text{GMU}_i}(\mathbf{w}, b)},$$

which can be written as

$$\mathcal{J}_{\text{GMU}}(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathbf{x} \sim f_{\mathbf{x}_i}} [\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b))], \quad (91)$$

where each expectation is taken over the per-sample GMM density $f_{\mathbf{x}_i}$. A natural question (and a natural sanity check) is: if we replaced each expectation by a Monte Carlo average over N IID draws from $f_{\mathbf{x}_i}$, how close would the resulting *standard* SVM come to the closed-form SVM-GMU? By the strong law of large numbers,

$$\frac{1}{N} \sum_{k=1}^N \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i^{(k)} + b)) \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \mathbb{E}_{\mathbf{x} \sim f_{\mathbf{x}_i}} [\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x} + b))], \quad (92)$$

so the empirical hinge loss of a standard SVM trained on the $nN = 6N$ sampled points (each labeled by its parent’s y_i) converges to eq. (91) pointwise in $(\mathbf{w}, b)$. Under the convexity of the objective established in section 20, the *minimizers* converge as well. This experiment verifies that prediction empirically on the `close_separable` dataset.

27.2 Experimental Procedure

The reference SVM-GMU is trained as in section 26, yielding the parameters in eq. (89) (denoted $\mathbf{w}_{\text{ref}}, b_{\text{ref}}$ below). For each value of $N \in \{1, 5, 25, 100, 500, 2000, 5000, 20000\}$ we repeat the following procedure.

1. For each of the $n = 6$ observed points, draw N IID samples from $f_{\mathbf{x}_i}$ by picking a component index $m \in \{1, \dots, M_i\}$ according to $\boldsymbol{\pi}_i$ and then drawing a point from $\mathcal{N}(\boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)})$.
2. Label each sample by its parent’s y_i .
3. Stack the $6N$ labeled samples into a dataset and train a *standard* linear SVM on it (i.e., invoke the estimator without any `sample_uncertainty`, which reduces to the usual hinge-loss objective). The regularization parameter $\lambda = 10^{-2}$ and random seed are the same as for the reference SVM-GMU.
4. Record the learned $(\mathbf{w}_N, b_N)$ and then compute three boundary-disagreement metrics against $(\mathbf{w}_{\text{ref}}, b_{\text{ref}})$.

27.3 Boundary-Disagreement Metrics

To quantify how close a learned boundary is to the reference SVM-GMU boundary, we use three scalar metrics:

- **Angle between normals:**

$$\theta_N = \arccos(\hat{\mathbf{w}}_N^T \hat{\mathbf{w}}_{\text{ref}}), \quad \hat{\mathbf{w}} = \frac{\mathbf{w}}{\|\mathbf{w}\|}, \quad (93)$$

reported in degrees. Measures how much the two boundary *orientations* disagree.

- **Offset difference:**

$$\Delta_N^{\text{offset}} = \left| \frac{b_N}{\|\mathbf{w}_N\|} - \frac{b_{\text{ref}}}{\|\mathbf{w}_{\text{ref}}\|} \right|. \quad (94)$$

Since $-\frac{b}{\|\mathbf{w}\|}$ is the signed distance from the origin to the hyperplane, this is the displacement of one boundary relative to the other along their shared normal direction.

- **Grid RMS:** the root-mean-square difference between the two *unit-normalized* decision functions, evaluated on a dense grid. Writing $\hat{f}(\mathbf{x}) = \hat{\mathbf{w}}^T \mathbf{x} + \frac{b}{\|\mathbf{w}\|}$ for the unit-normalized decision function of a boundary $(\mathbf{w}, b)$, so that $\hat{f}_N$ is this function built from the learned boundary $(\mathbf{w}_N, b_N)$ and $\hat{f}_{\text{ref}}$ is the same function built from the reference SVM-GMU boundary $(\mathbf{w}_{\text{ref}}, b_{\text{ref}})$, and letting $\{\mathbf{x}_g\}_{g=1}^G$ be the $G = 60 \times 60 = 3600$ points of a regular grid over the plotting window $[-5, 5] \times [-4, 5]$,

$$\Delta_N^{\text{grid}} = \sqrt{\frac{1}{G} \sum_{g=1}^G \left(\hat{f}_N(\mathbf{x}_g) - \hat{f}_{\text{ref}}(\mathbf{x}_g) \right)^2}. \quad (95)$$

In simple terms: because $\hat{f}(\mathbf{x})$ is exactly the signed distance from the point $\mathbf{x}$ to the boundary, this metric goes to every point of the window, asks how differently the two boundaries “see” that point (how far each one says the point is from it), squares those differences, averages them, and takes the square root. It is therefore the typical size of the disagreement between the two boundaries across the whole window, expressed as a single number. Unlike the angle and the offset, which isolate orientation and position separately, the grid RMS folds both effects into one overall summary; it is zero only when the two boundaries coincide everywhere in the window.

If the Monte Carlo argument above is correct, all three metrics should tend to zero as $N \rightarrow \infty$.

27.4 Results

Table 3 reports the three metrics for every value of N in the sweep, alongside the total training-set size $n_{\text{total}} = 6N$ used to fit each standard SVM. The trend is clear: by $N = 20000$ (i.e., $n_{\text{total}} = 120000$) the angle is within 0.22° , the offset difference is below 10^{-3} , and the grid RMS is ≈ 0.01 . Two minor non-monotonicities appear in the table (e.g., the angle rises slightly from $N = 25$ to $N = 100$), which we attribute to finite-sample noise and the stochastic nature of the SGD solver rather than to any systematic effect; the overall convergence is unmistakable.

Table 3: Convergence of the standard-SVM-on-samples boundary to the reference SVM-GMU boundary, as a function of the number of Monte Carlo draws N per observed point. $n_{\text{total}} = 6N$ is the size of the training set fed to the standard SVM.

N	n_{total}	θ_N (deg)	Δ_N^{offset}	Grid RMS
1	6	9.3425	1.0432	1.0917
5	30	4.2608	0.1404	0.2662
25	150	0.1996	0.1921	0.1912
100	600	1.2658	0.0463	0.0727
500	3000	0.4823	0.0231	0.0309
2000	12000	0.2334	0.0242	0.0280
5000	30000	0.2437	0.0032	0.0128
20000	120000	0.2168	0.0005	0.0106

Figure 4 plots the three metrics against N on a logarithmic horizontal axis. All three curves trend toward zero as N grows, which is exactly the pattern predicted by eq. (92) and the convexity of the objective.

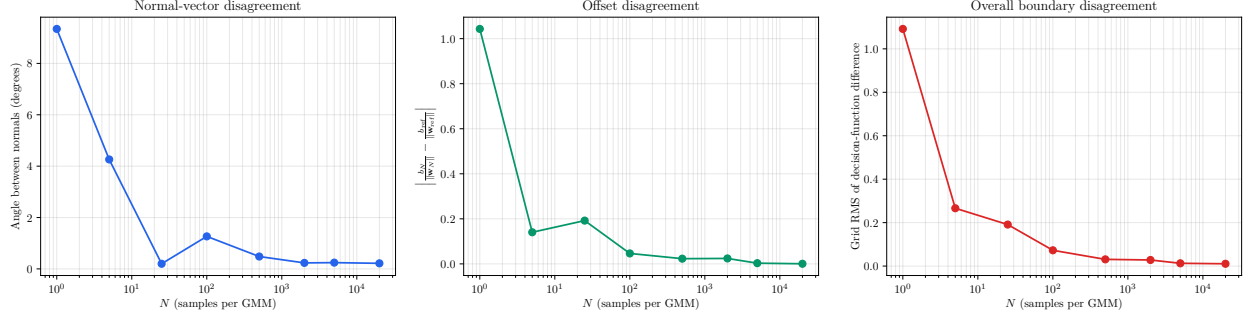


Figure 4: Boundary-disagreement metrics between the standard SVM fit to $6N$ Monte Carlo samples and the reference SVM-GMU, as a function of N . *Left*: angle between the two weight-vector normals (degrees). *Middle*: offset difference $\left| \frac{b_N}{\|\mathbf{w}_N\|} - \frac{b_{\text{ref}}}{\|\mathbf{w}_{\text{ref}}\|} \right|$. *Right*: RMS difference of the unit-normalized decision functions over a dense grid. All three tend to zero as N grows, confirming that the standard-SVM-on-samples minimizer converges to the SVM-GMU minimizer.

Figure 5 shows the visual counterpart of the metrics table: one panel per value of N , each depicting

1. the GMM contours of the original six observed points,
2. the sampled cloud used to fit that panel’s standard SVM (subsampled to at most 800 points for readability),
3. the reference SVM-GMU boundary as a dashed green line, and
4. the standard SVM boundary as a solid black line.

As N increases from 1 to 20000, the cloud fills out the GMM densities and the standard-SVM boundary pulls toward the SVM-GMU reference.

27.5 Interpretation

The experiment confirms, empirically, what section 14 argues on theoretical grounds: the SVM-GMU objective is nothing more than the $N \rightarrow \infty$ limit of a standard hinge-loss objective evaluated on samples drawn from the per-sample GMMs. In the opposite direction, it also highlights the computational value of the closed-form objective. Reaching a grid-RMS of ≈ 0.01 required $N = 20000$ samples per observed point, i.e., a training set of 120000 points. SVM-GMU attains the same fit directly from the $n = 6$ original examples and their GMM descriptions, with no sampling at all. In higher dimensions the gap between the closed-form and sampling approaches widens sharply (as noted in the concentration bound eq. (57)), so this advantage becomes essential rather than merely convenient.

28 Experiment 3: Does the Mixture Structure Matter?

28.1 Motivation

The SVM-GSU framework of section 9.5 already handles per-sample uncertainty via a single Gaussian per example. A natural question is: how much does the *mixture* structure of SVM-GMU actually buy us, beyond using the best single-Gaussian approximation of each mixture? We answer

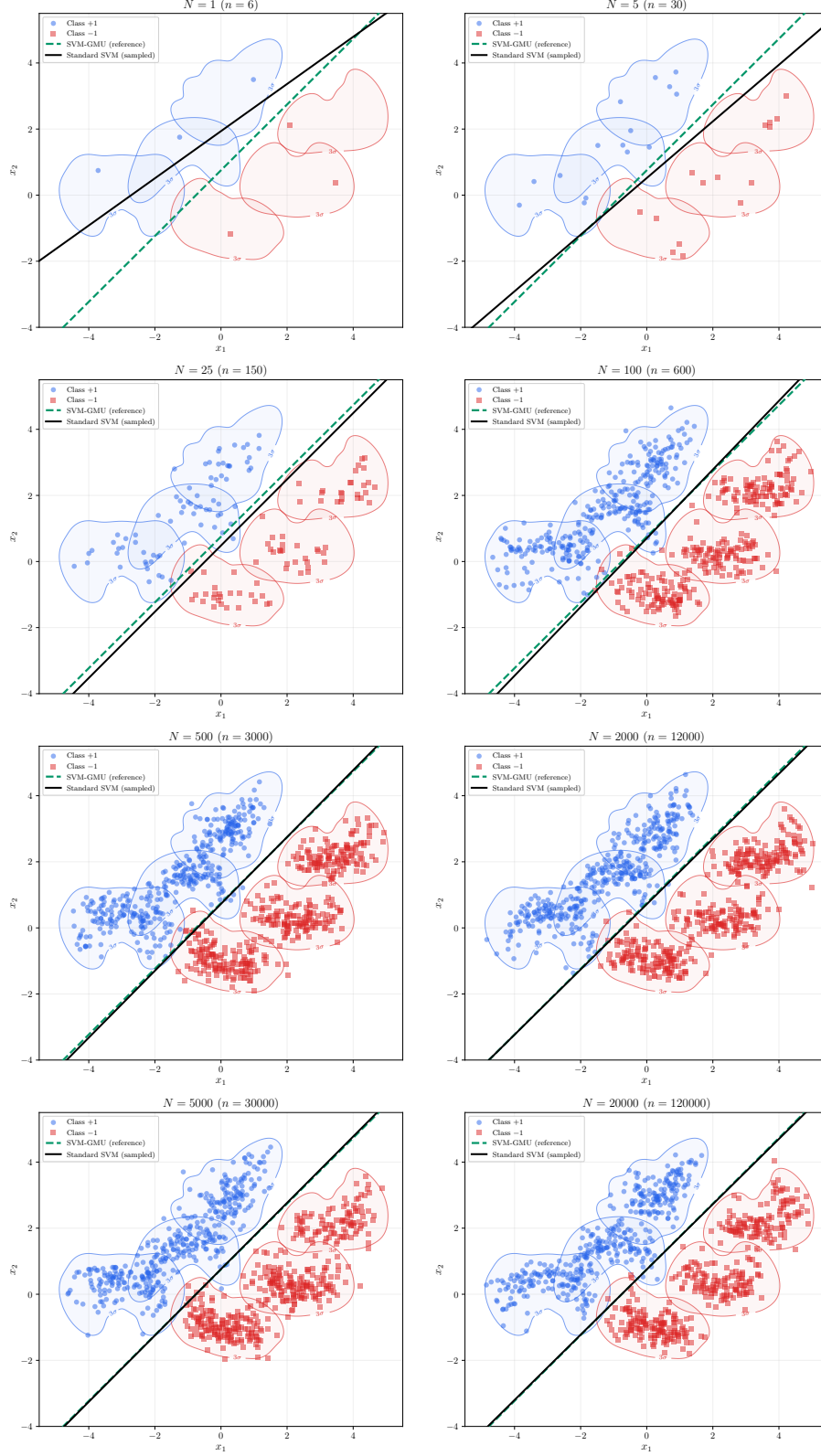


Figure 5: Convergence of the standard-SVM-on-samples boundary to the SVM-GMU boundary on the `close_separable` dataset.

this by comparing SVM-GMU against SVM-GSU on the `close_separable` dataset, where each GMM is replaced by a moment-matched single Gaussian.

28.2 Moment-Matched Single-Gaussian Approximation

For a GMM with weights $\{\pi_i^{(m)}\}$, component means $\{\boldsymbol{\mu}_i^{(m)}\}$, and component covariances $\{\boldsymbol{\Sigma}_i^{(m)}\}$ for $m = 1, \dots, M_i$, the moment-matched single Gaussian $\mathcal{N}(\bar{\boldsymbol{\mu}}_i, \bar{\boldsymbol{\Sigma}}_i)$ has the same overall mean and overall covariance as the mixture. The overall mean is the weighted average of component means,

$$\bar{\boldsymbol{\mu}}_i = \sum_{m=1}^{M_i} \pi_i^{(m)} \boldsymbol{\mu}_i^{(m)}, \quad (96)$$

and the overall covariance is given by the law of total covariance eq. (74), which we restate here in the notation of this experiment:

$$\bar{\boldsymbol{\Sigma}}_i = \underbrace{\sum_{m=1}^{M_i} \pi_i^{(m)} \boldsymbol{\Sigma}_i^{(m)}}_{\text{within-component (average component cov.)}} + \underbrace{\sum_{m=1}^{M_i} \pi_i^{(m)} (\boldsymbol{\mu}_i^{(m)} - \bar{\boldsymbol{\mu}}_i)(\boldsymbol{\mu}_i^{(m)} - \bar{\boldsymbol{\mu}}_i)^\top}_{\text{between-component (cov. of the means)}}. \quad (97)$$

The between-component term is the crucial one for banana- and crescent-shaped mixtures: it inflates the moment-matched covariance along the direction in which the component means spread out, turning a curved row of small ellipses into a single large, oriented ellipse.

Table 4 lists the moment-matched mean $\bar{\boldsymbol{\mu}}_i$ and the two eigenvalues of $\bar{\boldsymbol{\Sigma}}_i$ for each of the six samples, along with the anisotropy ratio λ_2/λ_1 (ratio of the larger to the smaller eigenvalue). The anisotropy is between $3.29\times$ and $3.99\times$, confirming that each moment-matched ellipse is heavily elongated, as one would expect from a banana/crescent GMM whose means lie along an arc.

Table 4: Moment-matched single-Gaussian approximations of the six GMMs in the `close_separable` dataset, obtained via eqs. (96) and (97). $\lambda_1 \leq \lambda_2$ are the eigenvalues of $\bar{\boldsymbol{\Sigma}}_i$. The anisotropy ratio λ_2/λ_1 is around 3 to 4 for every sample, which reflects the arc shape of the component means.

i	y_i	$\bar{\boldsymbol{\mu}}_i$	$\lambda_1(\bar{\boldsymbol{\Sigma}}_i)$	$\lambda_2(\bar{\boldsymbol{\Sigma}}_i)$	λ_2/λ_1
1	+1	$[-3.000 \quad +0.260]^\top$	0.188	0.672	3.57
2	+1	$[-1.064 \quad +1.388]^\top$	0.151	0.551	3.66
3	+1	$[+0.290 \quad +3.020]^\top$	0.149	0.508	3.40
4	-1	$[+0.142 \quad -0.944]^\top$	0.122	0.486	3.99
5	-1	$[+2.300 \quad +0.300]^\top$	0.144	0.474	3.29
6	-1	$[+3.509 \quad +2.214]^\top$	0.136	0.510	3.74

28.3 Training and Learned Parameters

Using the moment-matched Gaussians as a single-component GMM per sample (weight 1, mean $\bar{\boldsymbol{\mu}}_i$, covariance $\bar{\boldsymbol{\Sigma}}_i$), we fit SVM-GSU with the same hyperparameters as SVM-GMU ($\lambda = 10^{-2}$,

$T = 1000$, $k = 6$, seed 42). The learned parameters are:

$$\mathbf{w}_{\text{GMU}} = \begin{bmatrix} -1.2527 \\ 1.2589 \end{bmatrix}, \quad b_{\text{GMU}} = -0.9470, \quad (98)$$

$$\mathbf{w}_{\text{GSU}} = \begin{bmatrix} -1.1922 \\ 1.2721 \end{bmatrix}, \quad b_{\text{GSU}} = -0.9824. \quad (99)$$

Applying the three boundary-disagreement metrics from section 27 (here comparing the GSU boundary to the GMU boundary) gives:

$$\theta = 1.7157^\circ, \quad \Delta^{\text{offset}} = 0.0303, \quad \text{Grid RMS} = 0.0861. \quad (100)$$

The disagreement is small but not zero. For context: in the Monte Carlo experiment of section 27, the standard-SVM-on-samples boundary reaches a grid RMS of this magnitude (≈ 0.09) somewhere between $N = 25$ and $N = 100$ samples per observed point (i.e., $n_{\text{total}} \approx 150\text{--}600$ total training points, see table 3). So going from a full GMM to a single moment-matched Gaussian costs us about as much boundary fidelity as replacing the closed-form objective with a coarse Monte Carlo approximation.

28.4 Visual Comparison

Figure 6 shows the comparison in three panels. Panel (a) repeats the original GMM contours from fig. 1 for reference. Panel (b) shows the moment-matched single-Gaussian contours at the 3σ level. The loss of fidelity is immediate: each banana/crescent has collapsed into a single oriented ellipse that *covers* roughly the same region but assigns non-trivial probability mass to the interior of the original arc, exactly where the GMM placed almost no density. Panel (c) overlays the SVM-GMU boundary (dashed green) and the SVM-GSU boundary (solid purple) on top of the original GMM contours. The two boundaries differ by about 1.72° in orientation and by ≈ 0.03 in offset, with the SVM-GSU boundary pulled slightly toward the side where the moment-matched ellipses “spill over” more aggressively than the true GMM contours do.

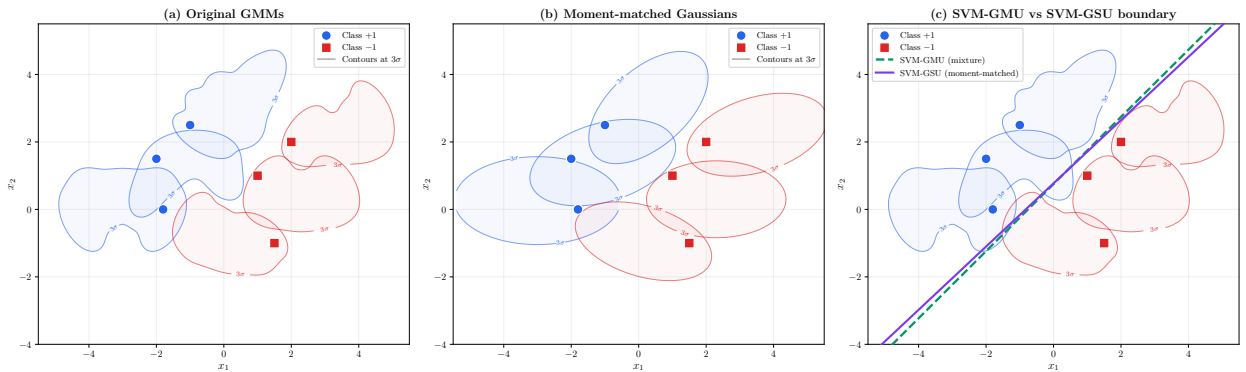


Figure 6: Effect of replacing the per-sample GMMs with their moment-matched single Gaussians. *Panel (a)*: original GMM contours (banana and crescent shapes). *Panel (b)*: moment-matched single-Gaussian contours at the same 3σ level; each GMM collapses to a tilted ellipse that covers roughly the same region but loses the shape information. *Panel (c)*: SVM-GMU boundary (dashed green) vs. SVM-GSU-on-moment-matched boundary (solid purple), overlaid on the original GMM contours. The two boundaries differ by the quantities in eq. (100).

28.5 Interpretation

Two observations stand out.

1. **Moment-matching is a lossy summary even when it preserves location and spread.** Panels (a) and (b) of fig. 6 have identical first and second moments *by construction*, yet they tell completely different stories about where probability mass actually lies. This is a fundamental limitation of any single-Gaussian representation of a genuinely non-elliptical uncertainty: the Gaussian family simply does not contain the shape.
2. **The shape loss propagates to the classifier.** Panel (c) shows that the two classifiers still end up close, but visibly distinct. On this dataset, with banana/crescent GMMs that are only moderately eccentric, the effect is small (grid RMS ≈ 0.086). On datasets where the per-sample uncertainty is more sharply non-Gaussian, or where samples lie closer to the decision boundary, one would expect the gap to grow accordingly.

The practical takeaway is the symmetric one we would want: when the per-sample uncertainty *is* unimodal and roughly elliptical, SVM-GSU and SVM-GMU will agree closely and the simpler GSU model is enough. When the uncertainty is multimodal, curved, or otherwise non-Gaussian, as in the `close_separable` dataset, SVM-GMU earns its keep by modeling the structure directly.

29 Experiment 4: Learning the Uncertainty from Data via Expectation Maximization

29.1 Motivation

The first three experiments assume the per-sample GMMs are handed to us. In a real application they are not: a sensor yields a cloud of noisy measurements, and the mixture that describes each example’s uncertainty must be *estimated*. This experiment closes that gap. For each observed point we draw samples from its true GMM, fit a fresh GMM to those samples by Expectation Maximization (EM), and run SVM-GMU on the estimated mixtures. Sweeping the number of samples shows how the learned boundary approaches the closed-form reference of section 26 as more data becomes available. It is the estimation-side counterpart of the Monte Carlo experiment of section 27: there we sampled and fit a standard SVM, here we sample, *re-learn the mixtures*, and fit SVM-GMU.

29.2 Experimental Procedure

The reference SVM-GMU is the one trained in section 26 on the true GMMs, with the parameters in eq. (89). For each value of $N \in \{50, 100, 250, 500, 1000, 2000, 5000, 20000\}$ we repeat the following.

1. For each of the $n = 6$ observed points, draw N IID samples from its true GMM $f_{\mathbf{x}_i}$ (component picked by $\boldsymbol{\pi}_i$, then a draw from that component’s Gaussian).
2. Fit a GMM to those N samples by EM, selecting the number of components $M \in \{1, \dots, 8\}$ that minimizes the Bayesian Information Criterion (BIC; see section 29.3), with full covariances and several restarts.
3. Label the estimated GMM by its parent’s y_i .

4. Fit SVM-GMU on the six estimated GMMs ($\lambda = 10^{-2}$, $T = 1000$, $k = 6$, seed 42) and measure its boundary against the reference using the three metrics of section 27 (angle between normals, offset difference, and grid RMS of the unit-normalized decision functions).

A single run is used per N with a fixed master seed, so the sweep is reproducible. As in section 27, single-run EM and the BIC component choice introduce some jitter, so the metrics are not perfectly monotonic.

29.3 Model Selection with the Bayesian Information Criterion

Step 2 of the procedure leaves one choice open: how many Gaussian components should the re-estimated mixture have? Too few and the fit cannot capture the banana or crescent shape; too many and the extra components merely chase sampling noise. We resolve this with the *Bayesian Information Criterion* (BIC), a standard model-selection score.

Definition. For a model fit by maximum likelihood to N data points, the BIC is

$$\text{BIC} = -2 \ln \hat{L} + k \ln N, \quad (101)$$

where $\hat{L}$ is the maximized likelihood of the data under the fitted model, k is the number of free parameters of the model, and N is the number of data points (here, the size of the sampled cloud for one observed example). The score has two competing terms:

- The *fit term* $-2 \ln \hat{L}$ rewards explaining the data. A better fit means a higher likelihood, which makes this term smaller (more negative). Adding components can only keep the likelihood the same or improve it, so this term decreases as M grows.
- The *complexity penalty* $k \ln N$ punishes using more parameters. It grows with M , since each additional component adds parameters.

Parameter count of a GMM. For a mixture of M Gaussians in d dimensions with full covariances, the free parameters are the $M - 1$ independent mixing weights (they sum to one), the Md entries of the means, and the $M(d(d+1)/2)$ distinct entries of the symmetric covariance matrices:

$$k = (M - 1) + Md + M \frac{d(d+1)}{2}. \quad (102)$$

In this experiment $d = 2$, so $k = 6M - 1$: every extra component costs six parameters, and the penalty $k \ln N$ rises accordingly.

Why the smallest BIC is chosen. Lower BIC is better. Selecting the M with the smallest BIC picks the model that best balances fit against complexity. Choosing by likelihood alone would always select the most complex candidate ($M = 8$ here), because more components never hurt the in-sample likelihood; the penalty $k \ln N$ is exactly what prevents that overfitting. Plotted against M , the BIC is typically U-shaped: it falls while added components yield genuine likelihood gains, then rises once further components only add parameters without improving the fit. The minimum is the sweet spot between underfitting and overfitting.

Consistency, and the trend in the M column. The penalty per parameter is $\ln N$, which *grows with the sample size*. This makes BIC consistent: as $N \rightarrow \infty$ it recovers the true number of components and resists adding spurious ones. This is exactly the behavior of the M column of table 5. At $N = 50$ the penalty dominates the modest likelihood gains that a few noisy points can demonstrate, so BIC collapses each example to roughly one component, $M = (1, 2, 1, 1, 1, 1)$. At $N = 20000$ the likelihood improvement from the genuine five to six components in each banana or crescent overwhelms the penalty, and BIC recovers counts near the truth, $M = (5, 4, 6, 4, 4, 4)$.

What BIC compares, and what it does not. It is worth being precise about the two distinct comparison steps in this experiment, because they operate at different levels.

1. **BIC (model selection, upstream).** For a single observed example, the N sampled points are fit by eight candidate GMMs, one for each component count $M \in \{1, \dots, 8\}$ (each with several EM restarts). BIC compares *these eight fitted mixtures* and keeps the one with the smallest score. This is done independently for each of the $n = 6$ observed examples, so per value of N there are $6 \times 8 = 48$ candidate fits and six selections; this is why M is reported as a six-vector rather than a single number.
2. **Boundary metrics (the measured outcome).** The two SVM-GMU boundaries, the true-GMM reference of section 26 and the one fit on the BIC-selected estimated GMMs, are compared by the three boundary-disagreement metrics of section 27 (angle, offset, grid RMS). BIC plays no role in that comparison.

In short, BIC only decides how many Gaussians to use when reconstructing each example’s uncertainty; it never sees the classifier. The convergence the experiment measures is then assessed separately by the boundary metrics.

29.4 Results

Table 5 reports the three metrics across the sweep, and figs. 7 and 8 visualize them. The trend is clear in the orientation and offset of the boundary: the angle falls to 0.26° and the offset difference to 7×10^{-4} by $N = 20000$, and the grid RMS ends at 0.0127, comparable to the 0.0106 reached by the Monte Carlo experiment of section 27 at the same N . The most telling quantity is the number of components BIC selects: at $N = 50$ it picks $M = (1, 2, 1, 1, 1, 1)$, collapsing most examples to a single Gaussian, whereas by $N = 20000$ it picks $M = (5, 4, 6, 4, 4, 4)$, approaching the true counts $(5, 6, 5, 6, 5, 6)$. As the data reveals the true mixture structure, the EM-fitted contours sharpen into the banana and crescent shapes and the learned boundary settles onto the reference.

Table 5: Disagreement between the SVM-GMU boundary fit on EM-estimated GMMs and the true-GMM reference, as a function of the number of EM training samples N per observed point ($n_{\text{total}} = 6N$). The column M lists the BIC-selected number of components for the six observed points ($M_1, \dots, M_6$); the true counts are (5, 6, 5, 6, 5, 6). A single run per N is used; the non-monotonicities (for example the $N = 100$ row) are single-run EM and BIC jitter, as also seen in table 3.

N	n_{total}	θ_N (deg)	Δ_N^{offset}	Grid RMS	M (BIC)
50	300	0.145	0.0049	0.0082	(1, 2, 1, 1, 1, 1)
100	600	2.287	0.1440	0.1708	(2, 2, 1, 1, 1, 1)
250	1500	0.792	0.0299	0.0519	(3, 2, 1, 2, 2, 2)
500	3000	0.383	0.0115	0.0208	(3, 3, 1, 2, 2, 3)
1000	6000	0.748	0.0067	0.0382	(3, 3, 2, 2, 3, 3)
2000	12000	0.858	0.0132	0.0457	(3, 3, 3, 2, 3, 3)
5000	30000	0.495	0.0031	0.0242	(5, 4, 5, 3, 3, 3)
20000	120000	0.260	0.0007	0.0127	(5, 4, 6, 4, 4, 4)

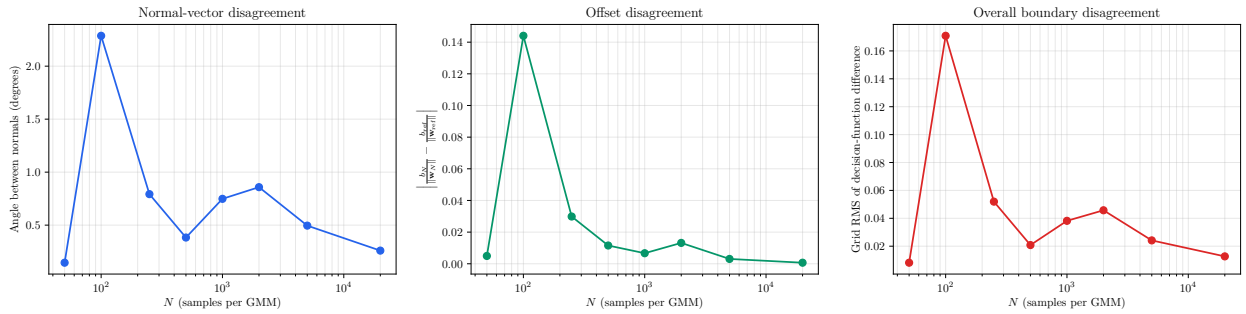


Figure 7: Boundary-disagreement metrics between SVM-GMU fit on EM-estimated GMMs and the true-GMM reference, versus the number of EM samples N per observed point. *Left*: angle between the weight-vector normals. *Middle*: offset difference. *Right*: grid RMS of the unit-normalized decision functions. The orientation and offset converge steadily toward zero; the grid RMS is noisier but ends near 0.013.

29.5 Interpretation

Two points stand out. First, SVM-GMU does not require the uncertainty to be known in closed form: when each mixture is estimated from data by EM with BIC choosing the component count, the resulting classifier converges to the one trained on the true mixtures. This is the bridge from the synthetic dataset to a realistic pipeline: collect repeated noisy measurements, fit a GMM per example, and feed those to SVM-GMU. Second, the convergence is not uniform across the metrics. The orientation and offset of the boundary settle quickly and cleanly, while the grid RMS is noisier and even dips deceptively low at $N = 50$. The reason is instructive: with very little data BIC underfits each example to a near-single Gaussian, which is close to the moment-matched single-Gaussian summary of section 28, and that summary already produces a boundary close to SVM-GMU. The meaningful convergence is therefore the trend at larger N , where the estimated mixtures genuinely recover the multi-component banana and crescent shapes and all three metrics

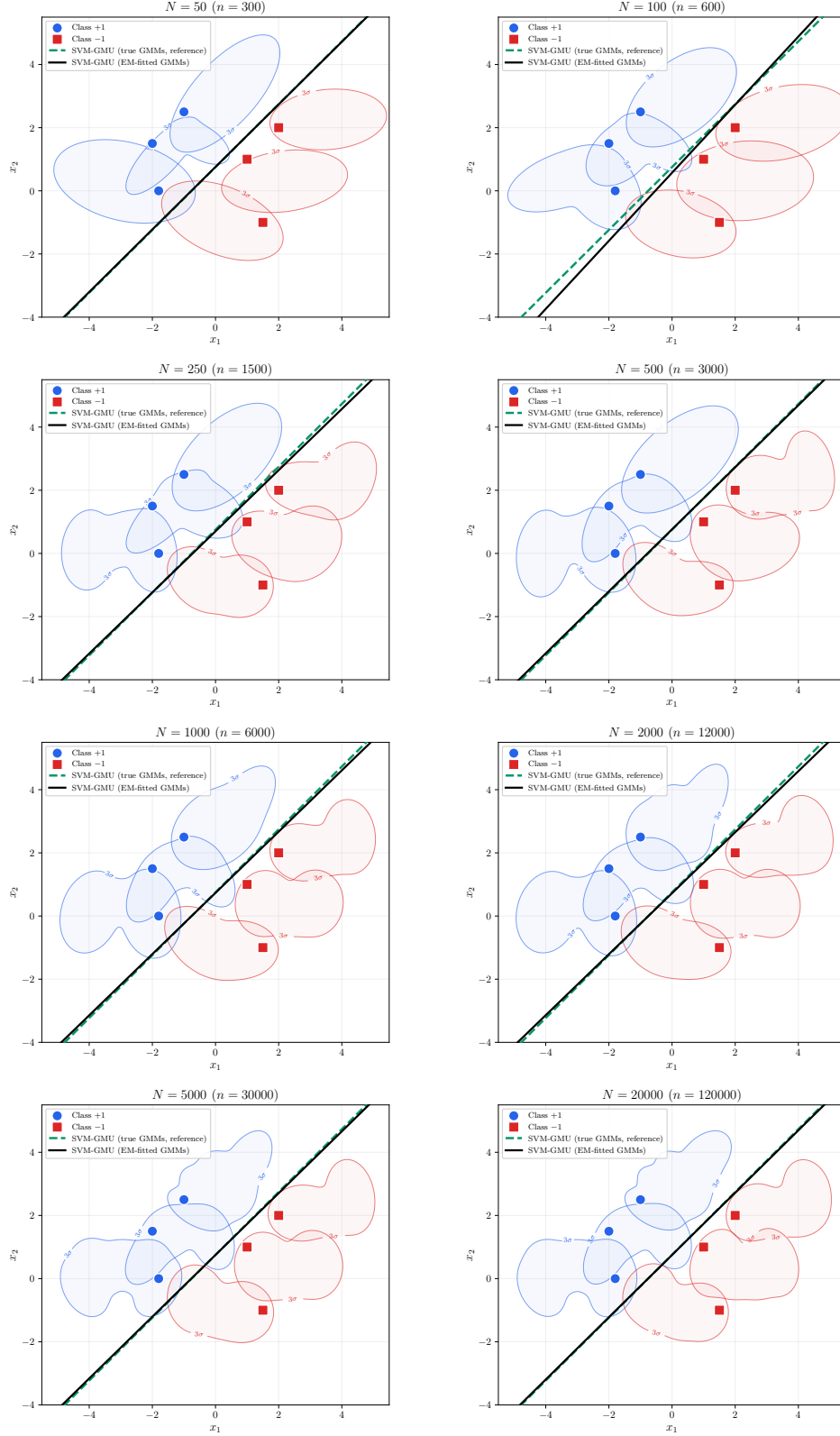


Figure 8: One panel per N : the EM-fitted GMM contours of the six points (at the 3σ level), the reference SVM-GMU boundary (green dashed), and the SVM-GMU boundary fit on the estimated GMMs (black solid). As N grows the fitted contours recover the true banana and crescent shapes and the black boundary approaches the green reference.

settle near zero.

Part VI

Discussion and Open Questions

30 How to Obtain the GMM Parameters in Practice?

The GMM parameters $\{\pi_i^{(m)}, \boldsymbol{\mu}_i^{(m)}, \boldsymbol{\Sigma}_i^{(m)}\}$ can be obtained from domain knowledge, by fitting a GMM to multiple measurements via the EM algorithm, from data augmentation, or from generative models with mixture posteriors.

31 Choosing the Number of Components M_i

Standard model selection criteria (BIC, AIC, cross-validation) apply. Even $M_i = 2$ or 3 dramatically increases expressiveness, as shown by the banana/crescent shapes in the `close_separable` experiments (Part V), which use $M_i \in \{5, 6\}$.

32 Potential for Kernelization

Kernelization remains open: the closed-form loss depends on $\mathbf{w}^\top \boldsymbol{\Sigma}_i^{(m)} \mathbf{w}$, which does not directly map to a kernel formulation.

33 Relationship to Other Non-Gaussian Uncertainty Models

Alternatives include sampling-based methods (which scale poorly, as the concentration bound eq. (57) and the convergence experiment of section 27 both show), robust optimization (more conservative), and other distributional families. GMMs offer the best flexibility due to their universal approximation property while preserving all computational advantages of the SVM-GSU framework.

Appendices

A Full Proof of theorem 9.1

We give a complete proof. The goal is to evaluate

$$I^+(\mathbf{a}, b') = \int_{\Omega^+} (\mathbf{a}^\top \mathbf{x} + b') f_{\mathbf{x}}(\mathbf{x}) \, d\mathbf{x}, \quad (103)$$

where $\Omega^+ = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}^\top \mathbf{x} + b' \geq 0\}$ and $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The case I^- follows by a symmetric argument.

We proceed by a sequence of four changes of variables, each bringing the integral closer to a standard form.

A.1 Step 1: Translation to Center the Gaussian

Let $\mathbf{y} = \mathbf{x} - \boldsymbol{\mu}$, so $\mathbf{x} = \mathbf{y} + \boldsymbol{\mu}$ and $d\mathbf{x} = d\mathbf{y}$. The Gaussian becomes

$$f_{\mathbf{x}}(\mathbf{y} + \boldsymbol{\mu}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2} \mathbf{y}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y}\right), \quad (104)$$

which is the density of $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. The integrand becomes

$$\mathbf{a}^\top (\mathbf{y} + \boldsymbol{\mu}) + b' = \mathbf{a}^\top \mathbf{y} + \underbrace{(\mathbf{a}^\top \boldsymbol{\mu} + b')}_{d_{\boldsymbol{\mu}}} = \mathbf{a}^\top \mathbf{y} + d_{\boldsymbol{\mu}}. \quad (105)$$

The halfspace transforms to

$$\Omega_1^+ = \{\mathbf{y} \in \mathbb{R}^d : \mathbf{a}^\top \mathbf{y} + d_{\boldsymbol{\mu}} \geq 0\}. \quad (106)$$

So

$$I^+ = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \int_{\Omega_1^+} (\mathbf{a}^\top \mathbf{y} + d_{\boldsymbol{\mu}}) \exp\left(-\frac{1}{2} \mathbf{y}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y}\right) d\mathbf{y}. \quad (107)$$

A.2 Step 2: Diagonalize the Covariance

Since $\boldsymbol{\Sigma} \in \mathbb{S}_{++}^d$, it has an orthonormal eigendecomposition:

$$\boldsymbol{\Sigma} = \mathbf{U} \mathbf{D} \mathbf{U}^\top, \quad (108)$$

where $\mathbf{U}$ is orthogonal ($\mathbf{U}^\top \mathbf{U} = \mathbf{I}$, so $|\det(\mathbf{U})| = 1$) and $\mathbf{D}$ is diagonal with positive entries (the eigenvalues of $\boldsymbol{\Sigma}$). Let $\mathbf{z} = \mathbf{U}^\top \mathbf{y}$, so $\mathbf{y} = \mathbf{U} \mathbf{z}$ and $d\mathbf{y} = |\det(\mathbf{U})| d\mathbf{z} = d\mathbf{z}$.

Define $\mathbf{a}_1 = \mathbf{U}^\top \mathbf{a}$. Then

$$\mathbf{a}^\top \mathbf{y} = \mathbf{a}^\top \mathbf{U} \mathbf{z} = (\mathbf{U}^\top \mathbf{a})^\top \mathbf{z} = \mathbf{a}_1^\top \mathbf{z}, \quad (109)$$

and

$$\mathbf{y}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y} = \mathbf{z}^\top \mathbf{U}^\top (\mathbf{U} \mathbf{D} \mathbf{U}^\top)^{-1} \mathbf{U} \mathbf{z} = \mathbf{z}^\top \mathbf{U}^\top \mathbf{U} \mathbf{D}^{-1} \mathbf{U}^\top \mathbf{U} \mathbf{z} = \mathbf{z}^\top \mathbf{D}^{-1} \mathbf{z}. \quad (110)$$

Also $|\boldsymbol{\Sigma}|^{1/2} = |\mathbf{D}|^{1/2}$. The integral becomes

$$I^+ = \frac{1}{(2\pi)^{d/2} |\mathbf{D}|^{1/2}} \int_{\Omega_2^+} (\mathbf{a}_1^\top \mathbf{z} + d_{\boldsymbol{\mu}}) \exp\left(-\frac{1}{2} \mathbf{z}^\top \mathbf{D}^{-1} \mathbf{z}\right) d\mathbf{z}, \quad (111)$$

where $\Omega_2^+ = \{\mathbf{z} : \mathbf{a}_1^\top \mathbf{z} + d_{\boldsymbol{\mu}} \geq 0\}$.

A.3 Step 3: Rescale to Standardize

Let $\mathbf{v} = \mathbf{D}^{-1/2} \mathbf{z}$, so $\mathbf{z} = \mathbf{D}^{1/2} \mathbf{v}$ and $d\mathbf{z} = |\mathbf{D}|^{1/2} d\mathbf{v}$. Define $\mathbf{a}_2 = \mathbf{D}^{1/2} \mathbf{a}_1$. Then

$$\mathbf{a}_1^\top \mathbf{z} = \mathbf{a}_1^\top \mathbf{D}^{1/2} \mathbf{v} = (\mathbf{D}^{1/2} \mathbf{a}_1)^\top \mathbf{v} = \mathbf{a}_2^\top \mathbf{v}, \quad (112)$$

and

$$\mathbf{z}^\top \mathbf{D}^{-1} \mathbf{z} = \mathbf{v}^\top \mathbf{D}^{1/2} \mathbf{D}^{-1} \mathbf{D}^{1/2} \mathbf{v} = \mathbf{v}^\top \mathbf{v}. \quad (113)$$

The factors $|\mathbf{D}|^{1/2}$ cancel, yielding

$$I^+ = \frac{1}{(2\pi)^{d/2}} \int_{\Omega_3^+} (\mathbf{a}_2^\top \mathbf{v} + d_{\boldsymbol{\mu}}) \exp\left(-\frac{1}{2} \|\mathbf{v}\|^2\right) d\mathbf{v}, \quad (114)$$

where $\Omega_3^+ = \{\mathbf{v} : \mathbf{a}_2^\top \mathbf{v} + d_{\boldsymbol{\mu}} \geq 0\}$.

Note that

$$\|\mathbf{a}_2\|^2 = \mathbf{a}_2^\top \mathbf{a}_2 = \mathbf{a}_1^\top \mathbf{D} \mathbf{a}_1 = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{U}^\top \mathbf{a} = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}, \quad (115)$$

so $\|\mathbf{a}_2\| = \sqrt{\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}}$. Recall from eq. (23) that $d_{\boldsymbol{\Sigma}} = \sqrt{2\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}} = \sqrt{2} \|\mathbf{a}_2\|$.

A.4 Step 4: Rotation to Isolate One Coordinate

Now we have a standard Gaussian (with identity covariance) and a linear function times it. Pick an orthogonal matrix $\mathbf{B}$ such that

$$\mathbf{B}\mathbf{a}_2 = \|\mathbf{a}_2\|\mathbf{e}_d, \quad (116)$$

where $\mathbf{e}_d = (0, 0, \dots, 0, 1)^\top$ is the d -th standard basis vector. Such a $\mathbf{B}$ exists (for example, a Householder reflection).

Let $\mathbf{m} = \mathbf{B}\mathbf{v}$, so $\mathbf{v} = \mathbf{B}^\top\mathbf{m}$ and $d\mathbf{v} = d\mathbf{m}$. Then

$$\mathbf{a}_2^\top \mathbf{v} = \mathbf{a}_2^\top \mathbf{B}^\top \mathbf{m} = (\mathbf{B}\mathbf{a}_2)^\top \mathbf{m} = \|\mathbf{a}_2\| \mathbf{e}_d^\top \mathbf{m} = \|\mathbf{a}_2\| m_d, \quad (117)$$

and

$$\|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v} = \mathbf{m}^\top \mathbf{B} \mathbf{B}^\top \mathbf{m} = \mathbf{m}^\top \mathbf{m} = \|\mathbf{m}\|^2. \quad (118)$$

The halfspace becomes $\Omega_4^+ = \{\mathbf{m} : \|\mathbf{a}_2\| m_d + d_\mu \geq 0\}$, equivalently $m_d \geq -d_\mu / \|\mathbf{a}_2\|$.

So eq. (114) becomes

$$I^+ = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^{d-1}} \int_{-d_\mu / \|\mathbf{a}_2\|}^{\infty} (\|\mathbf{a}_2\| m_d + d_\mu) \exp\left(-\frac{1}{2} \|\mathbf{m}\|^2\right) dm_d dm_1 \cdots dm_{d-1}. \quad (119)$$

A.5 Step 5: Separate and Evaluate the 1D Integral

Write $\|\mathbf{m}\|^2 = m_d^2 + \sum_{j=1}^{d-1} m_j^2$. The integrand factors and the $m_1, \dots, m_{d-1}$ integrals each give $\sqrt{2\pi}$ (they are integrals of a standard Gaussian over all of $\mathbb{R}$):

$$\int_{\mathbb{R}^{d-1}} \exp\left(-\frac{1}{2} \sum_{j=1}^{d-1} m_j^2\right) \prod_{j=1}^{d-1} dm_j = (2\pi)^{(d-1)/2}. \quad (120)$$

So

$$I^+ = \frac{(2\pi)^{(d-1)/2}}{(2\pi)^{d/2}} \int_c^\infty (\|\mathbf{a}_2\| t + d_\mu) \exp\left(-\frac{t^2}{2}\right) dt = \frac{1}{\sqrt{2\pi}} \int_c^\infty (\|\mathbf{a}_2\| t + d_\mu) \exp\left(-\frac{t^2}{2}\right) dt, \quad (121)$$

where $c = -d_\mu / \|\mathbf{a}_2\|$. We split this into two 1D integrals:

$$I^+ = \frac{\|\mathbf{a}_2\|}{\sqrt{2\pi}} \underbrace{\int_c^\infty t e^{-t^2/2} dt}_{T_1} + \frac{d_\mu}{\sqrt{2\pi}} \underbrace{\int_c^\infty e^{-t^2/2} dt}_{T_2}. \quad (122)$$

Evaluating T_1 . Since $\frac{d}{dt}(-e^{-t^2/2}) = t e^{-t^2/2}$:

$$T_1 = \left[-e^{-t^2/2}\right]_c^\infty = 0 - (-e^{-c^2/2}) = e^{-c^2/2}. \quad (123)$$

Evaluating T_2 . Using the substitution $u = t/\sqrt{2}$, $dt = \sqrt{2} du$:

$$T_2 = \int_c^\infty e^{-t^2/2} dt = \sqrt{2} \int_{c/\sqrt{2}}^\infty e^{-u^2} du. \quad (124)$$

Now we express this in terms of the error function. From eq. (24), $\int_0^x e^{-u^2} du = \frac{\sqrt{\pi}}{2} \text{erf}(x)$, and $\int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2}$. Therefore

$$\int_{c/\sqrt{2}}^\infty e^{-u^2} du = \int_0^\infty e^{-u^2} du - \int_0^{c/\sqrt{2}} e^{-u^2} du = \frac{\sqrt{\pi}}{2} - \frac{\sqrt{\pi}}{2} \text{erf}\left(\frac{c}{\sqrt{2}}\right) = \frac{\sqrt{\pi}}{2} \left(1 - \text{erf}\left(\frac{c}{\sqrt{2}}\right)\right). \quad (125)$$

So

$$T_2 = \sqrt{2} \cdot \frac{\sqrt{\pi}}{2} \left(1 - \text{erf}\left(\frac{c}{\sqrt{2}}\right)\right) = \sqrt{\frac{\pi}{2}} \left(1 - \text{erf}\left(\frac{c}{\sqrt{2}}\right)\right). \quad (126)$$

A.6 Step 6: Combine and Simplify

Putting it together:

$$I^+ = \frac{\|\mathbf{a}_2\|}{\sqrt{2\pi}} e^{-c^2/2} + \frac{d_\mu}{\sqrt{2\pi}} \cdot \sqrt{\frac{\pi}{2}} \left(1 - \text{erf}\left(\frac{c}{\sqrt{2}}\right)\right). \quad (127)$$

Simplify the second term: $\frac{1}{\sqrt{2\pi}} \cdot \sqrt{\frac{\pi}{2}} = \frac{1}{2}$. So

$$I^+ = \frac{\|\mathbf{a}_2\|}{\sqrt{2\pi}} e^{-c^2/2} + \frac{d_\mu}{2} \left(1 - \text{erf}\left(\frac{c}{\sqrt{2}}\right)\right). \quad (128)$$

Now substitute back. Recall $c = -d_\mu/\|\mathbf{a}_2\|$, and $\|\mathbf{a}_2\| = d_\Sigma/\sqrt{2}$, so $c = -\sqrt{2} d_\mu/d_\Sigma$, and

$$\frac{c}{\sqrt{2}} = -\frac{d_\mu}{d_\Sigma}, \quad \frac{c^2}{2} = \frac{d_\mu^2}{d_\Sigma^2}. \quad (129)$$

Using the oddness of erf, $\text{erf}(-x) = -\text{erf}(x)$:

$$1 - \text{erf}\left(-\frac{d_\mu}{d_\Sigma}\right) = 1 + \text{erf}\left(\frac{d_\mu}{d_\Sigma}\right). \quad (130)$$

Also

$$\frac{\|\mathbf{a}_2\|}{\sqrt{2\pi}} = \frac{d_\Sigma/\sqrt{2}}{\sqrt{2\pi}} = \frac{d_\Sigma}{2\sqrt{\pi}}. \quad (131)$$

Therefore

$$I^+ = \frac{d_\mu}{2} \left(1 + \text{erf}\left(\frac{d_\mu}{d_\Sigma}\right)\right) + \frac{d_\Sigma}{2\sqrt{\pi}} \exp\left(-\frac{d_\mu^2}{d_\Sigma^2}\right). \quad (132)$$

This is exactly eq. (21) with the + sign. The I^- case follows by repeating the argument with the halfspace $\mathbf{a}^\top \mathbf{x} + b' \leq 0$; equivalently, one can use the identity $I^+ + I^- = \mathbb{E}[\mathbf{a}^\top \mathbf{x} + b'] = d_\mu$ to recover I^- . $\square$

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